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▶ **To cite this version:**

Shirin Rafiei, Chetna Singhal, Kjell Brunnström, Jonas Andersson, Mårten Sjöström. Laboratory study on quality of experience and user experience for teleoperation. Quality and User Experience, 2026, 11 (1), pp.1. ⟨10.1007/s41233-025-00076-3⟩. ⟨hal-05548309⟩

HAL Id: hal-05548309

https://hal.science/hal-05548309v1

Submitted on 25 Mar 2026

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Quality and User Experience (2026) 11:1
https://doi.org/10.1007/s41233-025-00076-3

ORIGINAL PAPER

Laboratory study on quality of experience and user experience for teleoperation

Shirin Rafiei[1,2] · Chetna Singhal[3] · Kjell Brunnström[1,2] · Jonas Andersson[1] · Mårten Sjöström[2]

Received: 7 March 2025 / Accepted: 30 November 2025
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Abstract
This study explores how video quality, field of view, and latency influence users' performance, depth perception, and overall experience in remote-control systems for industrial teleoperated applications. In this controlled laboratory investigation, we conducted an experimental study using a test setup that closely replicated real-world conditions while ensuring experimental control and safety. Participants completed 18 trials under varying fields of view (Wide vs. Narrow), latency (High, Medium, Low), and video quality (High, Medium, Low) conditions to evaluate their navigation accuracy, depth perception and Quality of Experience. Using a mixed method approach, considering Quality of Experience and User Experience insights, we integrated objective performance metrics, quantitative user evaluations, and qualitative feedback to understand how visual constraints impact control and decision-making. The results show that latency had the most impact on navigation accuracy, followed by field of view. The high latency level significantly influenced performance, causing larger deviations from the intended path and reducing confidence in estimating depth. The effect was even more pronounced in narrow fields of view, where high latency further amplified navigation difficulties. In contrast, a wider field of view helped mitigate latency effects by improving situational awareness and reducing reliance on secondary visual cues. While video quality had a minimal impact on objective navigation performance, qualitative feedback indicated that higher video quality improved confidence in depth perception and reduced visual strain, supporting users in making precise judgments. We also collected users' feedback that provided cues to further enhance the teleoperation systems.

Keywords Remote control systems · Depth perception · Visual perception · Latency · Video quality · Field of view · Situational awareness · Industrial teleportation · Quality of experience · User experience · Augmented reality · Mixed-methods · User-centred design

Introduction

The rapid development of Remote Control Systems (RCSs) has revolutionised industries by enabling operators to manage Unmanned Ground and Aerial Vehicles (UGV/UAV) from safe, remote locations. While these systems contribute to the digital transformation central to Industry 4.0 [1], their emphasis on human–machine collaboration, user-centred design, and cognitive support aligns more closely with the vision of Industry 5.0. This next phase prioritises human well-being, ethical automation, and societal value, offering benefits such as enhanced workplace safety, reduced emissions, and improved productivity. Sectors such as mining [2, 3], forestry [4], and construction are increasingly adopting RCS technologies to improve operational efficiency and worker safety [5].

RCS and teleoperation are related concepts, but they differ in complexity and application. While RCS typically involves simple, direct, and line-of-sight interactions, teleoperation extends these principles to enable complex, high-precision operations in remote or challenging environments through advanced feedback and communication mechanisms. In this study, we target teleoperation, as it offers greater precision and situational awareness capabilities, making it more suitable for our research objectives [6, 7].

✉ Shirin Rafiei
shirin.rafiei@ri.se

1 RISE Research Institutes of Sweden, Stockholm, Sweden

2 Department of Computer and Electrical Engineering, Mid Sweden University, Sundsvall, Sweden

3 Inria, Rennes, France

Published online: 09 March 2026

Teleoperation tasks in automated vehicle systems generally fall into three categories: monitoring, assistance, and direct control [8]. Monitoring involves supervising autonomous operations and responding to system alerts, while assistance refers to intervening in specific situations where automation encounters uncertainty or limitations. Within direct control, a key challenge is delivering visual information to operators, as it is crucial for maintaining situational awareness and depth perception during complex tasks [9]. Situational awareness, as broadly defined by Endsley [10], encompasses the perception of environmental elements, comprehension of their meaning, and projection of their future status. While this definition encompasses cognitive and decision-making processes, this study focuses specifically on the visual component of situational awareness. This interpretation aligns with research on teleoperation, where maintaining a clear and continuous visual understanding of the workspace is essential for effective task execution [11, 12].

While teleoperation offers significant advantages, challenges such as network instability, low video quality, and system latency can negatively impact operator performance and decision-making [13–15]. These challenges underscore the importance of understanding user perceptions and experiences to enhance system usability and effectiveness [16–18]. Existing research has explored the impact of individual system components on user performance, such as video latency [4, 19], video quality [20, 21], and field of view [22]. However, only a limited number of studies have comprehensively investigated how the interplay between these factors affects user experience and depth perception in teleoperation [5, 23, 24].

The complex interplay between technological parameters, the user experience, and contextual understanding requires a user-centred, mixed-methods approach that integrates qualitative and quantitative evaluations. Quantitative methods measure task performance and system controllability, whereas qualitative methods capture operators' perspectives, contributing to a comprehensive assessment of system usability and potential areas for enhancement.

This study examines the impact of video quality (VQ), latency (Lat), and field of view (FoV) on depth perception, performance, and overall user experience in teleoperation systems. Combining subjective users' feedback with objective data aims to provide a deeper understanding of the user experience while interacting with teleoperation systems. Our specific contributions are as follows:

- *Assessment of user performance:* We quantify the impact of VQ, FoV, and latency on task precision and error rates in remote operations.
- *Evaluation of depth perception:* We analyse how variations in VQ, FoV, and latency influence operators' ability to accurately perceive depth in remote operation tasks.
- *Analysis of overall user experience:* We assess users' experience, interaction, and comfort through a combined User experience (UX) and Quality of Experience (QoE) framework to identify areas for system improvement.
- *Application of the UX and QoE framework:* We employ a holistic, user-centred approach that combines qualitative and quantitative evaluations to provide a comprehensive understanding of the interplay between users, technology, and context in teleoperation.

The remainder of this paper is organised as follows: "Background and related works" reviews related work and identifies existing gaps. "Method" details the experimental design and methodology. "Result" presents the experimental results and analysis. "Discussion" discusses the implications of the findings. Finally, "Conclusion" concludes the paper and outlines directions for future research.

Background and related works

Teleoperation enhances safety and efficiency, but significant challenges have been identified. Dadhich et al. [3], emphasised the advantages of reducing exposure to hazardous environments by allowing operators to control machinery from a distance. However, their study also identifies challenges such as depth perception, user experience, and task performance. Operators often struggle with accurately judging distances through 2D camera feeds, which can impact the precision and speed of their tasks. Improving these aspects is essential for maximising the benefits of teleoperation and ensuring remote operations. Table 1 outlines key factors influencing teleoperation performance and experience, helping to situate this study's focus–latency, video quality, and field of view–within the broader research landscape. This section explores advancements, challenges, and user experience considerations central to teleoperation, setting the foundation for the contributions of this work.

Factors influencing remote control operation

Various factors affecting human performance in teleoperation were examined by Chen et al. [27], which focused on user interface design. By reviewing and analysing existing studies on remote control interfaces, they identified key performance issues and best practices to enhance operator efficiency and accuracy. Their research highlighted that user interface elements, such as latency and field of view,

Table 1 Key factors influencing performance and experience in teleoperation systems

Category	Example Factors	Influence on Teleoperation
System/Technical	Latency, video quality, field of view, display mode, network conditions	Affects system responsiveness, visual clarity, and control accuracy.
Human/Cognitive	Depth perception, cognitive load, situational awareness, user comfort	Impacts perception, task efficiency, and decision-making.
Interface/Design	Control layout, stereoscopic vs monoscopic view, multisensory feedback	Shapes usability, immersion, and mental workload.
Task & Contextual	Task complexity, level of automation, operating environment	Determines information needs, interface demands, and control strategies.

Categories summarised from literature (e.g., [5, 19, 25, 26])

could be critical in teleoperation and significantly impact performance.

Latency

Communication latency is a critical challenge in teleoperation, as it can increase the cognitive load, reduce situational awareness, and increase task completion times and error rates [4, 19, 28]. Latency in teleoperation is influenced by several factors, including network conditions (e.g., bandwidth, packet loss, and jitter), communication latencies (time-varying and asymmetric delays), control algorithms (adaptive and predictive control), and system design (data compression and advanced network protocols). A study by Parrot et al. [29] examined various components contributing to latency to identify the key factors behind "Glass-to-Glass" latency and to design more efficient and responsive systems. The study revealed that, with advancements in network technology providing faster speeds, application-layer delays have become more significant than network delays. These delays are caused primarily by processing time, data complexity, encoding, and synchronisation overhead. Equipment limitations and computational delays can further increase latency, highlighting the need to optimise both software processes and hardware resources to improve system performance and responsiveness.

Video quality

Video quality plays a critical role in how effectively operators can interpret remote environments. Low video quality can compromise situational awareness and task performance, resulting in operator frustration and inefficiency. Research [30] has shown that improving video resolution can significantly enhance user comfort and performance in teleoperation tasks.

Field of view

The field of view of the visual feed is another crucial factor influencing the operator's ability to navigate and interact with the environment. Depth perception and spatial awareness are critical for effective task performance in teleoperation. Systems that offer multiple perspectives, such as First Person View (FPV) and Third Person View (TPV), enhance the UX by reducing the cognitive load. Additionally, optimising the field of view is essential, as a limited FoV can restrict awareness, whereas an overly wide FoV may cause distortion or overload. Tailored settings for view modes and FoVs can significantly improve accuracy and usability in teleoperation [25, 31].

Monoscopic vs stereoscopic presentation

Remote operation systems are often equipped with monocular camera systems because of their simplicity, lower cost, and ease of integration. For instance, commercial drones commonly use a single forward-facing camera to provide users with an FPV. While these systems are sufficient for basic navigation and operations, they may lack the depth perception necessary for complex tasks such as precise object manipulation. To address the limitations of monocular systems, teleoperation systems can be equipped with binocular cameras to provide stereoscopic representations, offering enhanced depth perception compared with monoscopic (2D) presentations [31, 32]. This capability is particularly beneficial for tasks requiring precise manipulation and navigation, such as robot teleoperation or rescue missions [33].

Studies have shown that stereoscopic displays can lead to faster task completion times and improved accuracy, as they allow operators to better understand spatial relationships in remote environments, especially in dynamic scenarios [34, 35]. Despite these advantages (e.g., faster completion time and improved accuracy), stereoscopic systems also present challenges. One significant issue is the vergence-accommodation conflict, which can impact operator performance [36–38]. Additionally, incorrect calibration of stereoscopic cameras can create difficulties in image fusion, leading to discomfort and reduced effectiveness of depth perception [32]. Last, the cost and complexity of stereoscopic systems, which often require specialised equipment such as binocular cameras and 3D displays, can be prohibitive [36, 37].

Therefore, while stereoscopic systems offer benefits in enhancing depth perception, they also come with notable challenges that need to be addressed. In this article, we will concentrate on the monoscopic presentations for teleoperation.

Depth perception

Depth perception poses a significant challenge in the remote control of cranes, as it impacts both task performance and operator safety. Sitompul et al. [39] investigated this issue by exploring various graphical user interfaces, including shadow and perspective line interfaces, designed to enhance depth perception. Their findings demonstrated that these interfaces improve spatial awareness and task performance, emphasising the importance of effective depth cues in remote operations. This research highlights depth perception as a critical factor in remote control systems and offers valuable insights for future interface design.

Maintaining situational awareness is another challenge in teleoperation, often complicated by the reliance on indirect visual and sensory feedback. Various approaches, such as Augmented Reality (AR) and multisensory inputs, have been developed to address this issue [12, 40, 41]. AR and multisensory inputs, such as haptics, enhance situational awareness by providing enriched feedback that captures the complexity of remote environments [11]. These advancements have contributed to overcoming limitations in situational awareness and supporting operators in navigating complex, remote environments effectively.

Integration of the QoE and UX in teleoperation

Understanding users' satisfaction and interaction in remote control systems is crucial for enhancing system performance, improving content delivery, and addressing human factors [42]. Evaluating how users perceive their experience and identifying ways to enhance their interaction can provide valuable insights for designing more effective and user-centred systems. In this context, two key concepts emerge: UX and QoE, which help assess the user's perceptions, expectations, and overall satisfaction.

UX is defined as "users' perceptions and responses that result from the use and/or anticipated use of a system, product, or service [43]." Similarly, QoE is defined as "the degree of delight or annoyance of the user of an application or service. It results from the fulfilment of his or her expectations concerning the utility and/or enjoyment of the application or service in the light of the user's personality and current state [44–46]."

While UX and QoE originate from different research traditions, Human–Computer Interaction (HCI) and Telecommunications, respectively, they are not opposing concepts. Both address the perceptual, emotional, and experiential aspects of user interaction, and their scopes have increasingly converged in recent literature [23, 24, 42, 46–48]. In this study, we distinguish between UX and QoE for methodological clarity, while analysing them jointly within a QUX framework. UX is explored through open-ended qualitative feedback, capturing reflective, experiential, and contextual aspects of interaction. QoE is operationalised via subjective quality ratings that reflect users' perceptual and affective responses to system conditions. Task performance, measured through objective metrics, is treated as a separate yet complementary evaluation dimension. While these metrics show behavioural outcomes, we also used user ratings and qualitative reflections, particularly on system responsiveness, controllability, and usability, to support their interpretation (as shown later in Table 5). This integration supports analytical clarity and reflects disciplinary conventions, aligning with ITU-T definitions [23, 48, 49], which acknowledge that QoE encompasses emotional, behavioural, and contextual factors traditionally associated with UX. Thus, while UX and QoE are treated separately in our analysis, they are understood as overlapping and complementary perspectives on the teleoperation experience.

Hassenzahl's framework [26, 50] for understanding user experience highlights three key dimensions: pragmatic, hedonic, and eudaimonic aspects. Pragmatic aspects focus on the functional and utilitarian qualities of a product, emphasising its ability to help users achieve specific goals effectively and efficiently. These are tied to "do-goals" and include attributes such as usability, reliability, and task performance. Hedonic aspects, on the other hand, emphasise the experiential and emotional qualities of a product, addressing "be-goals" such as stimulation, self-expression, and emotional connection. These qualities enhance user enjoyment by fulfilling deeper psychological needs, such as novelty, personalisation, and memory evocation. Finally, eudaimonic aspects focus on creating meaningful experiences that contribute to personal growth, self-realisation, and long-term well-being. These aspects address "fulfilment goals" by aligning products with users' intrinsic values and aspirations, fostering a sense of purpose and achievement. Together, these dimensions provide a comprehensive framework for designing products that are not only functional but also emotionally engaging and personally meaningful.

QoE research typically adheres to standardised procedures for measurements and tools, whereas UX methods are more diverse, drawing inspiration from multiple disciplines. Another distinction lies in the consideration of time spans. UX research often incorporates temporal aspects of user interactions, which have traditionally been less emphasised in QoE studies. However, recent efforts in the QoE

community have adopted user-centric approaches similar to those in UX to better capture QoE in practice [47]. These efforts highlight the importance of positive emotions, such as delight and enjoyment, which are associated with satisfying hedonic needs and enhancing user satisfaction. Despite these advancements, operationalising a fully user-centred QoE framework remains challenging, with UX considered more mature in its application and development.

Building on these dimensions, effective holistic evaluation in remote operation systems requires combining objective data (e.g., behavioural metrics) with subjective feedback (e.g., user self-reports) to capture both measurable and experiential aspects of user interaction [51]. This approach bridges the pragmatic and hedonic qualities outlined in Hassenzahl's framework while advancing the integration of the QoE and UX (QUX). The interdisciplinary QUX framework enhances usability and operational effectiveness. Remote vehicle operations research could benefit from the QUX approach, where aligning functional efficiency with meaningful and satisfying experiences is critical for addressing human factors and optimising system design. This aligns with the proposed framework for evaluating QoE in a testbed-oriented living lab setting [52], which emphasises the importance of both objective and subjective methods for capturing users' experiences in real-life settings.

Experimental control and real-world applicability

In RCSs, experimental studies are essential for evaluating system performance, user experience, and interaction under controlled conditions. These studies allow researchers to isolate specific variables, measure their effects precisely, and gain scientific insights into critical factors such as latency, user satisfaction, and task efficiency. However, a key challenge lies in ensuring that the findings from these controlled experiments are applicable to real-world scenarios, where environments and user interactions are far more complex and dynamic. While traditional laboratory experiments prioritise internal validity–ensuring that observed results stem from well-controlled conditions–they often sacrifice real-world applicability, making it harder to generalise findings beyond the laboratory setting. For teleoperation, where operations occur in diverse and unpredictable environments, balancing experimental control with practical relevance is particularly critical to ensure that the results can inform real-world system design and operation.

To address the challenge of translating controlled experimental findings into real-world applications, experimental designs should reflect the complexities of real-world conditions, including realistic settings, natural user behaviours, and dynamic tasks. Subramanian et al. [53] identify three dimensions necessary for achieving practical relevance: setting, users, and methodology. This includes designing experimental environments that mimic real-world tasks, selecting participants who represent actual operators, and adopting methodologies such as immersive or gamified scenarios to simulate authentic interactions [53].

Janowski et al. [54] further emphasised the importance of providing participants with greater freedom to interact with systems during experiments. This flexibility captures natural behaviours often overlooked in rigidly controlled studies. For example, incorporating real-world platforms such as YouTube into lab experiments not only enhances practical relevance but also generates richer data on user preferences and behaviours [54].

Finding a balance between experimental control and real-world applicability often requires hybrid approaches, such as combining lab-based studies with field tests or crowd-sourced evaluations. For teleoperation, this balance is essential for designing systems that are not only scientifically validated but also practical, reliable, and user-friendly in operational environments [55]. In this study, we address this balance through the lens of ecological validity (see Sections "Experiment procedure and data collection" and "Limitations of the Study"), ensuring that controlled lab conditions were grounded in realistic operational scenarios.

Knowledge gap

While previous studies have identified different challenges in teleoperation applications, a comprehensive understanding of how different factors, specifically video configuration of visual information, influence the overall experience of remote operators remains incomplete.

This study investigates the impact of VQ, FoV, and Lat on operator performance, depth perception, and the user experience during teleoperation. A lab-based remote control platform (LRCP) simulates real-world applications, such as controlling wheel loaders in construction tasks [3], and provides a controlled environment for evaluating user interactions. The controlled laboratory setting allows precise measurement of variables such as video degradation and latency, ensuring rigour before validating findings through field studies to enhance ecological relevance [56, 57]. To conceptually structure this investigation, the study adopts a QUX framework, integrating QoE and task performance as complementary dimensions. QoE is assessed via subjective quality ratings, UX through open-ended experiential feedback, and task performance via objective metrics. This structure enables a holistic, triangulated evaluation of teleoperation experience. The following research questions guide the study:

RQ. 1 What is the impact of VQ, FoV, and Latency on

users' performance in controlling and precisely navigating the teleoperation?

RQ. 2 What is the effect of VQ and FoV on depth perception in teleoperation?

RQ. 3 What is the influence of VQ, FoV, and Latency on the overall UX and QoE in teleoperation?

RQ. 4 How does a mixed-methodology QUX approach, combining qualitative and quantitative data, enhance the understanding of user interaction in teleoperation?

To answer these questions, the study employs the QUX framework, which integrates subjective evaluations (e.g., user feedback) and objective metrics (e.g., task performance) to provide a holistic perspective on user interaction [58][47]. Building on prior work that assessed depth judgment tasks across various interface configurations [30,59], this study employs a triangulation design (see Sect. 3) to investigate how video configurations impact teleoperation users.

Method

This section describes how the research questions outlined in the previous section are addressed. A triangulated mixed-method approach was employed to examine the effects of different visual configurations in teleoperation on task performance, depth perception, and user experience. Triangulation is a research strategy that enhances validity and reliability by integrating multiple data sources, methods, or perspectives [60]. Common in mixed-methods research, it integrates quantitative data with qualitative insights to provide a more comprehensive analysis and reduce biases, thereby strengthening the credibility of the research.

The research design consists of an experimental study and a qualitative evaluation. The experimental study investigates user performance and depth perception under varying visual conditions, leveraging objective performance metrics, subjective QoE evaluations, and user feedback (RQ1-3). The quantitative and qualitative evaluations together assess the methodological approach used in RQ1-3, ensuring that the research framework effectively captures QUX (RQ4).

This section details the study design, methodology, and data collection process, including the experimental setup, participant recruitment, system apparatus, analysis techniques, and ethical considerations.

Study design

The experiment was conducted at a RISE lab in Stockholm, Sweden, using an LRCP designed to mimic tasks in industrial teleoperation applications. The user-centred lab-based approach was employed in developing the LRCP, which was inspired by discussions with Volvo Construction Equipment to address real-world challenges in teleoperation. Figure 1 illustrates the test setup.

This study examined the impact of VQ, FoV, and system latency on remote operators' task performance, depth perception, and overall user experience when controlling wheel loaders in construction scenarios. Due to the complexity and safety risks of testing these factors in actual environments [3], a low-scale simulation using a remote-controlled toy truck was developed.

A pilot study with four participants was conducted to evaluate the experimental setup, environment, questionnaires, and procedures. This initial test identified key issues in the study design and informed refinements to improve the accuracy and reliability of the main experiment. The original setup included a straight navigation path and four numerically labelled target objects (Fig. 4, left), but pilot feedback revealed that the labels introduced unintended depth cues and that the path did not adequately assess navigation performance under varying latency conditions. In response, the main experiment was modified to use three coloured objects (red, yellow, and green; Fig. 4, right) to eliminate size-related depth cues and to include a longer, curved track, enhancing both ecological validity and control over visual configuration effects.

In the study, participants interacted with the LRCP under varying video configurations. Their feedback was collected through a recurring questionnaire (explained more in Sec. 3.3) using the Absolute Category Rating (ACR) method, as defined by ITU-T Recommendations P.910 [61], to assess perceived quality. To complement these structured

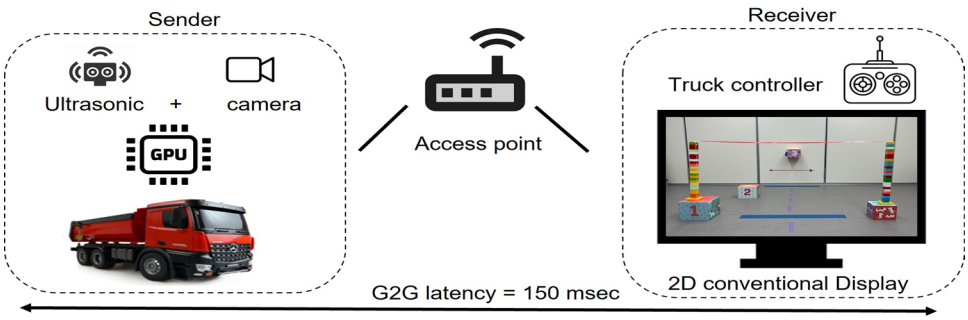


Fig. 1 Test setup including sender (Remote-site) and receiver (On-site). The right-hand side of this image shows the participant's perspective from inside the truck

measures, open-ended questions were included to capture more nuanced insights into participants' experiences, revealing observations and challenges that predefined metrics might overlook.

Apparatus

Experiment setup

The experimental setup consisted of a remote-controlled toy truck (Amewi 22407), an off-the-shelf camera, and a dedicated WiFi network. The toy truck served as a scaled-down model to replicate remote vehicle teleoperation scenarios, with control provided via a game controller (Fig. 1). A Basler Dart daA1600-60uc camera with a fixed C-Mount CCTV lens is mounted on the truck to offer an FPV, and it connects via USB 3.0 to an NVIDIA Jetson Nano GPU for real-time video processing.

We implemented a video streaming pipeline using GStreamer [62], a flexible media framework that enables the chaining of different media elements to multiplex and create advanced media pipelines. The designed GStreamer pipeline (sender side) captures video from a Basler camera (`pylonsrc`), rescales it to 640×480 resolution at 30 frames per second, compresses it as a video stream using the `x264enc` encoder [62] with a target bitrate of 30 Mbps (`bitrate=30000`), a `superfast` speed preset to prioritize encoding speed, and a minimum quantization parameter (`qp-min=30`) to control compression quality. The compressed stream is packetized with `rtph264pay` and transmitted over UDP. On the receiver side, another GStreamer pipeline depayloads the stream (`rtph264depay`), decodes it (`decodebin`), and displays it on a 2D conventional LCD display. The LRCP system is designed for ultra-low latency real-time video streaming, with a baseline Glass-to-Glass latency (LoLat) of less than 150 ms [63, 64].

A GoPro Hero 8 camera was mounted on the ceiling to record the toy truck's navigation during the trials, providing both validation and objective data analysis. The footage was used to track the truck's position and movement, as detailed in "Objective data".

To evaluate the effects of varying video streaming conditions on user performance, depth perception, and overall user experience, three factors were systematically manipulated: video quality, latency, and field of view. Video quality was adjusted by modifying the video resolution parameters while maintaining a constant frame rate. To assess the impact of increased latency, controlled buffering delays were introduced into the GStreamer pipeline: the medium latency condition (MedLat) added 250 ms for a total latency of 400 ms, and the high latency condition (HiLat) added 500 ms, resulting in a total latency of 650 ms. FoV variations were implemented by integrating a cropping mechanism within the GStreamer sender pipeline. This adjustment reduced the FoV from 120 degrees (WideFoV) to 105 degrees (NarrowFoV) by cropping the video at the source. To ensure consistent presentation, the cropped image was rescaled to match the original display size. This approach preserved the visual proportions while maintaining a clear representation for the operator, ensuring that changes in FoV did not affect the perceived visual size on the display. However, this rescaling involved interpolating non-captured pixels using the default bilinear interpolation method, which may introduce mild blur. The perceptibility of this blur can vary based on the operator's viewing distance from the display. We considered a fixed viewing distance for all users during the experiment. The experimental settings are summarised in Table 2.

To enhance the ecological validity of the mimicked RC setup, the chosen toy truck's speed range was between 2.8 - 3.3 m/s (about 10 km/h), similar to the speed of industrial construction machines in real-world ToP use cases. Indeed, the ambient illumination levels and colour temperatures were adjusted to approximate outdoor conditions on a cloudy day. The lighting was calibrated to D65 (6500 K), ensuring the colour temperature closely matched natural daylight. However, the standard laboratory lighting was insufficient to replicate the intensity of outdoor summer conditions.

Table 2 Test Configuration Conditions: Video Quality, Latency, and Field of View (FoV)

Resolution levels (Frame rate 30 fps)	Latency (ms)	FoV (degrees)
HiQual: 720 × 640	LoLat: 150	Wide-FoV: 120
MedQual: 640 × 480	MedLat: 400	Narrow-FoV: 105
LoQual: 480 × 320	HiLat: 650	–

FoV= Field of View. *HiQual*= High Quality; LoLat = Low Latency

Latency measurement methods

Since latency is a key element for dependent factors in this study, we employed precise latency measurement methods to control our trial settings. These measurements were critical for analysing the influence of latency on the dependent variables and ensuring the reliability of the experimental results. The various latency components involved in the Glass-to-Glass latency of the LRCP and the corresponding evaluation methods are summarised in Table 3.

Glass-to-Glass latency: Figure 2 illustrates the two methods used for Glass-to-Glass latency measurement, i.e., on-screen and flasher pulse timer methods. Figure 3 shows the average values of the latency measured using these methods for the video quality levels that were used in the trials.

Table 3 Types of Latency Involved and Evaluation Methods

No.	Latency Component	Evaluation Method	Average (ms)	SD (ms)
1	Glass-to-glass	On-screen capture	136	5
		Flasher–pulse timer	142	12
2	Handheld controller	Counter	34	12
3	Network transmission	Network analyser	30	10
4	Display lag	Leo Bodnar latency tester	11	6

SD Standard deviation. Glass-to-glass latency was measured using two different methods

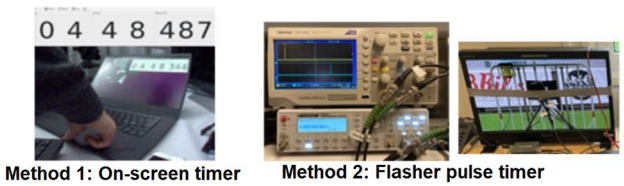


Fig. 2 Glass-to-Glass latency evaluation methods

On-screen timer: In this latency measurement method, a camera is placed in front of the receiver's display. The display showed a timer and streamed the video from the camera simultaneously, allowing for an accurate latency assessment. The camera captured the view, and the video stream passed through the entire pipeline. We then took a snapshot of the screen showing the timer and the streamed video output on a media player window (Fig. 2). The media player window on the screen shows the timer value with the latency compared to the actual timer visible on the same screen. This latency corresponds to the Glass-to-Glass latency between the source (a visually acquired scene using a camera) and the receiver's display.

Flasher pulse timer We also used a measurement setup for absolute latency developed by Brunnström et al. [65] to capture Glass-to-Glass latency. This method measures the accurate latency using a white flash. The in-frame latency of the flash and its time duration is chosen to fit within only one frame of the video stream. A counter is used to measure latency, which directly shows the time delay, and this measurement is also read by a computer.

Remote control latency: Time taken between the remote control user input and the corresponding response on the truck is done using a high frame rate camera that captures the remote control input trigger (user hand movement), the toy truck movement, and a counter (1 ms resolution), in the same video. The game controller latency, measured over ten trials, was observed to be less than 40 ms.

Network transmission latency: We have used the Wireshark tool to capture network traces and evaluate the network latency. The TCP-based round-trip time on average is 80 ms, and the standard deviation is 25 ms, while the UDP-based time delta on average is 30 ms, and the standard deviation is 10 ms.

Display lag: We used the Leo Bodnar Video Signal 4 K Latency Tester to measure the video signal latency from its input until it is displayed on the monitor. An HDMI video signal is sent from the measurement instrument to the display. This will present rectangular areas on the screen. When the measurement instrument's optical sensor is placed on a rectangular area, the time delay for the signal to reach that area is displayed on the same screen. The input latency measurement accuracy is in milliseconds. We used Lenovo ThinkVision P27h-10, 27". The measurements reported a latency of 1.2 ms for the top of the monitor, 8.6 ms for the middle, and 16.2 ms for the bottom. As the screen was run in 60 Hz, i.e., 16.7 ms, the screen did not add any measurable latency.

Experiment procedure and data collection

The experiment varied the Visual Configurations Under Test (ATUT), with participants performing the same tasks using the LRCP system under different conditions. To minimise

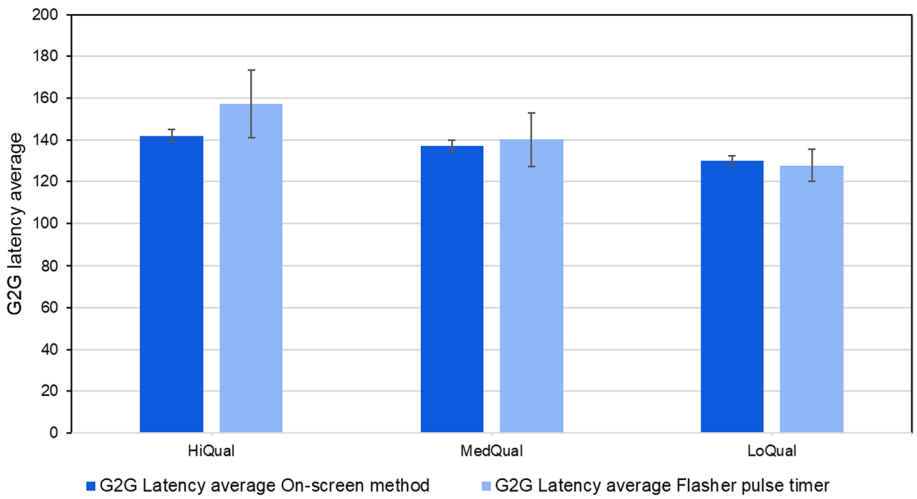


Fig. 3 Glass-to-Glass latency average for the On-screen method in High Video Quality, Medium Video Quality, and Low Video Quality compared to the Flasher method

order effects and reduce learning or fatigue-related biases, the presentation order of visual configurations was counterbalanced using a pseudo-randomised scheme. Each participant experienced all eighteen conditions in a predefined sequence, and the order was recorded.

Upon arrival, participants were welcomed and introduced to the LRCP system. They signed a consent form [61] and read the experiment instructions, which outlined the tasks and the interaction with the ATUT. Participants were informed that they would complete eighteen trials under different visual conditions, but no further details were provided about the nature of these conditions.

Before the driving training session, participants attended a video quality training session to familiarise themselves with the range of video conditions they might encounter in the main experiment. A 50 cm reference scale was also marked on the site floor to support depth perception tasks (Fig. 4). Following this, participants completed a hands-on training session using the default ATUT configuration, during which they practised teleoperation tasks with a game controller. They received a verbal overview of the remote control system and were encouraged to explore the interface to gain a deeper understanding of its functionality.

Each trial required participants to complete specific tasks (e.g., depth perception and navigation) using a designated visual setup. To assess user experience and performance, data were collected through five questionnaires distributed at different stages of the experiment via Google Forms:

Before the test: A background questionnaire and the Simulator Sickness Questionnaire (SSQ) [66] were used. The SSQ was used since the study aimed to capture visual and cognitive discomfort that can arise in teleoperation contexts. Although the setup was not fully immersive, factors such as visual latency, low video quality, and limited FoV can induce oculomotor strain. This approach aligns with research showing that even non-immersive systems may lead to simulator sickness [67], offering a baseline for future comparisons with stereoscopic or VR-based systems.

During the test: A recurring questionnaire after each trial based on the ACR method to assess user experience.

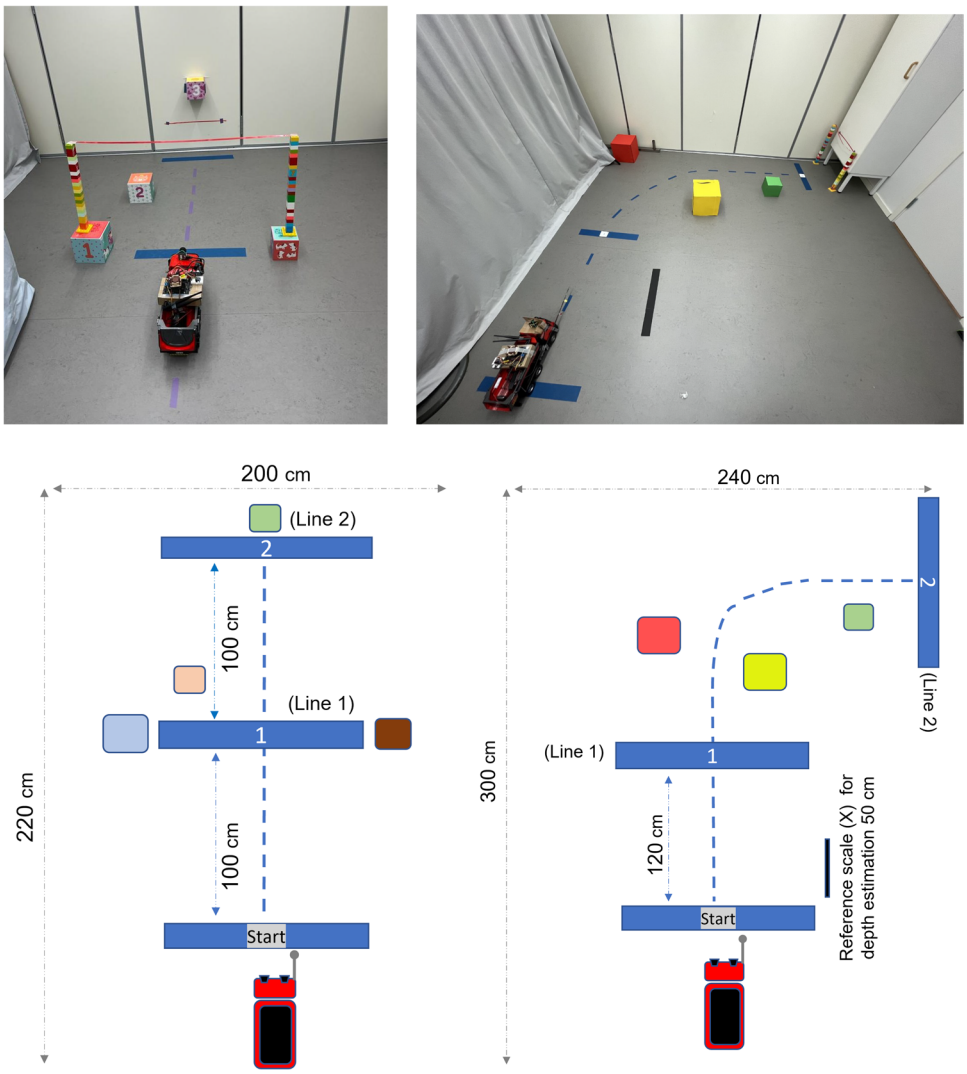


Fig. 4 Test setup for the pilot study (left) and main experiment (right). Based on pilot feedback, labels were removed to avoid depth cues. The main test used three unlabeled boxes: red (Object 1), yellow (Object 2), and green (Object 3)

After the test: A final SSQ and an open-ended feedback questionnaire to collect qualitative insights on user perceptions and challenges.

The recurring tasks consisted of three sections:

1. Distance Estimation: Participants estimated the distances to specified targets (Object 1: red box, Object 2: yellow box) placed in the test environment (Fig. 4 - right).
2. Navigation Task 1: Participants remotely navigated the LRCP from the start line and stopped before line 1, pausing for three seconds without crossing the line.
3. Navigation Task 2: Participants resumed driving from line 1 to line 2, stopping just before the line without touching it, followed by completing the recurring questionnaire.

The total time of all trials ranged between 1 to 1.5 h, with opportunities for mini-breaks between trials. The duration varied depending on participants' thoroughness, as some took more time to complete tasks with greater accuracy, while others proceeded more quickly. Although the total duration of each experimental session exceeded 30 min (average duration: 75 min), mini-breaks of approximately 2–3 min were given after every four task conditions (trials). These pauses were used to reduce cognitive strain and mitigate potential fatigue effects, in recognition of ITU-T P.910 guidelines [61]. All participants were encouraged to rest, move around if needed, and ask questions during these breaks before resuming the next set of trials.

Participants were compensated with a gift card valued at 100 SEK from RISE, regardless of whether they completed the experiment or not.

Additional items assessing perceived system responsiveness, usability, comfort, and controllability were custom-designed for this study based on established concepts in UX and QoE research [5, 68]. While these additional items are not part of a standardised instrument, they were developed to capture task-relevant perceptual feedback and triangulate findings from objective measures and open-ended responses.

Participants

Twenty naive participants took part in the experiment with 18 different randomised test runs (Trials) that are shown in Table 4.

The participants were recruited through advertising on the Accindi platform [69], a platform for finding individuals who volunteer as test subjects for studies. The 18 experimental trials were arranged using a pseudo-randomised counterbalancing scheme to vary the order of visual interface

Table 4 Counterbalanced test configurations across 18 trials combining latency (Lat), field of view (FoV), and video quality (VQ). High video quality (H-VQ) indicates the best resolution, while High Latency (H-Lat) represents the maximum added delay

Trial	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16	T17	T18
VQ	H	M	L	H	M	L	H	M	L	H	M	L	H	M	L	H	M	L
Lat	Low (L)			Medium (M)			High (H)			Low (L)			Medium (M)			High (H)		
FoV	Wide									Narrow								

H = High; M = Medium; L = Low; VQ = Video quality; Lat = Latency; FoV = Field of view. H-Qual indicates the best resolution; H-Lat represents the maximum added delay

Table 5 Research questions mapped to data collection

RQs	Objective data	User ratings	Qualitative reflections
RQ1: Performance	Truck deviation from the defined track	System responsiveness	System responsiveness
		Controllability of the system	System usability
			Multi-sensory impacts
RQ2: Depth perception	Average depth estimation deviation (ADED)	Impact of FoV on depth perception	Impact of FoV on depth perception
		Impact of VQ on depth perception	Impact of VQ on depth perception
		Users' confidence in depth perception	
RQ3: General UX		Users' comfort	System improvement
		Interaction experience rating (IER)	General remote experience (GRE)
		Simulator sickness	

conditions (VQ, FoV, and Latency) across participants. Although a formal Latin square was not used, a pseudo-randomisation scheme ensured that each visual condition appeared approximately equally across trial positions, helping to reduce order effects and learning biases.

The background questionnaire results reveal a diverse sample of participants in terms of age, experience, and other characteristics. Most participants (60%) were aged between 26–35 years, with smaller percentages in other age brackets, indicating a relatively young sample overall. The majority of participants (55%) reported no prior experience in piloting remote-controlled vehicles, while 45% had relevant experience (e.g., direct driving of remote-controlled toy cars), indicating a mixed level of familiarity with the task. Regarding eyewear, 65% of participants did not use any corrective lenses, while 3% wore glasses and 5% used lenses. The gender distribution was somewhat skewed, with 65% male and 35% female.

Ethical consideration

The test instructions emphasised the voluntary nature of participation. Participants signed an informed consent form and were informed of their right to take breaks or withdraw from the test at any time without providing a reason. All collected data were handled anonymously and confidentially, in full compliance with the General Data Protection Regulation (GDPR). Participants were informed that the study's focus was on testing the LRCP and its various visual interfaces, rather than evaluating the participants themselves.

Participant recruitment was primarily conducted through the Accindi platform, while for the pilot test, participants were invited from RISE students and staff. Staff participation was scheduled outside business hours. To minimise the collection of personal information, age was recorded in 10-year intervals to maintain anonymity while enabling analysis of potential age-related effects.

Data analysis

This section describes the data analysis methods used to evaluate performance, user experience, and perceived quality. A mixed-method approach integrated objective measures, subjective ratings, and qualitative feedback to address the research questions. To capture experiential dimensions such as responsiveness, comfort, usability, and controllability, we developed context-specific questionnaire items informed by established UX and QoE constructs. Although these items do not originate from standardised instruments, they supported data triangulation and a deeper interpretation of user experience in the teleoperation setting.

Data analysis - task performance (RQ1)

Task performance was analysed using a combination of objective data, user ratings, and users' qualitative reflections (shown in Table 5).

Objective data

Truck Deviation from Defined Track: Quantifies the deviation of the LRCP from the predefined navigation path, providing insights into accuracy and control during the task.

The user performance measurement setup is illustrated in Fig. 5. It consists of three operations: track position estimate, LRCP position estimate, and user performance metric computation. We discuss these points in further detail below.

The track position was identified using binary colour thresholding (Fig. 6), and the LRCP's position was estimated using the Speeded Up Robust Features (SURF) based feature matching [70]. The performance metric was calculated as the Mean Absolute Deviation (MAD) between the LRCP (Fig. 7) and the track.

Track position estimate: During each trial, the scene, S, is acquired using the Basler camera and the track, to be followed by the LRCP, is marked in the scene. Each pixel, $S(i, j)$, at location (i, j) in the scene, is represented by a three element column matrix, consisting of the RGB (red green

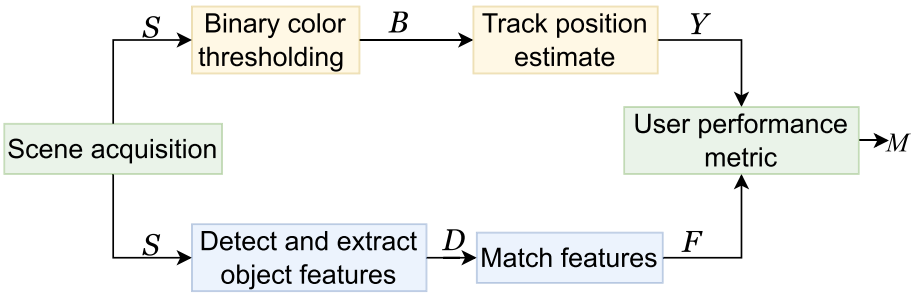


Fig. 5 User performance measurement setup: The scene (S) is processed to estimate track position (Y) and LRCP position (F). Performance metric (M) is the mean absolute difference between matched features and track points



Fig. 6 Track position estimate: Binary image B, highlighting the main pattern of the track that the truck should follow

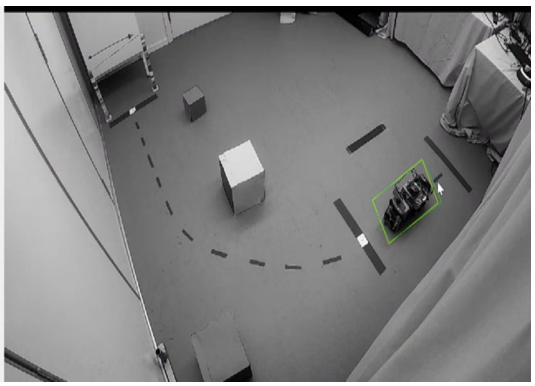


Fig. 7 LRCP position estimate: Tracking the movement of the LRCP

blue) color components, $[s_r(i,j), s_g(i,j), s_b(i,j)]^\intercal$. The track colour, C^T, is also represented by a similar three-element column matrix, $[c_r^T, c_g^T, c_b^T]^\intercal$. We apply binary colour thresholding to get the binary image output, B, as shown in Fig. 6. Each pixel of B at location (i,j) is defined as follows:

$$B(i,j) = \begin{cases} 1 \text{ if } |S(i,j) - C^T| \leq C^{th} \\ 0 \text{ otherwise} \end{cases} \quad (1)$$

where, $C^{th} = [c_r^{th}, c_g^{th}, c_b^{th}]^\intercal$ is the threshold deviation in colour components considered in the scene due to illumination and intensity variations during acquisition. The track position estimate is determined using the binary color threshold output, B, and it is defined as a set, Y, containing the pixel positions containing the track markings, $Y = \{(i,j) \,|\, B(i,j) = 1\}$.

LRCP position estimate: We have created a dataset of 50 images of the LRCP captured from all angles, as it might appear in the scene during the trials. We detect the SURF [70] from the LRCP (reference) and the scene images. We extract 100 strongest point features from both the reference and scene SURF sets, denoting them as D. We detect the LRCP in the scene by finding point correspondences (feature matching) between the reference (LRCP image) and the scene.

This approach is robust to scale change, in-plane (and out-of-plane) rotation, and occlusion. The matched feature set, F, contains the inlier matched points, i.e. the LRCP position estimate, in the scene image.

The bounding box polygon of the detected LRCP in the scene image, using F, is shown in Fig. 7.

User performance metric. We define the user performance metric, M using Mean absolute difference between the matched feature points $(i,j) \in F$ and the closest track point $(\tilde{i}, \tilde{j}) \in Y$, i.e.

$$M = \frac{\sum\limits_{\forall (i,j) \in F} (|i - \tilde{i}| + |j - \tilde{j}|)}{\mathcal{F}}, \mathcal{F} \geq 1 \quad (2)$$

$$(\tilde{i}, \tilde{j}) = \min_{(i',j') \in Y} \sqrt{(i-i')^2 + (j-j')^2}), \forall (i,j) \in F \quad (3)$$

where, $\mathcal{F}$ is the cardinality of the matched feature point set F. Ideally, if the LRCP position is on track, the value of M will be 0. Hence, a smaller value of this metric is desired, indicating a lower deviation of the LRCP from the track.

To evaluate the effects of latency, field of view, and video quality on users' performance, we applied the Aligned Rank Transform ANOVA (ART ANOVA) [71] on the performance metric. ART ANOVA is a non-parametric method that allows factorial analysis of variance while accommodating violations of normality and non-additive interactions, making it suitable for our dataset, which did not meet the assumptions of normality and sphericity. This method transforms the data by aligning it before ranking, ensuring that

Table 6 Formulated questions concerning recurring questionnaire, asked after each test run

Q.No	Questions asked from participants after each trial in open-ended questionnaire
Q.1	How would you rate your confidence in your estimated depth in the previous questions? (1=Bad; 2=Poor; 3=Fair; 4=Good; 5=Excellent)
Q.2	How would you rate your ability in estimating depth? (1=Bad; 2=Poor; 3=Fair; 4=Good; 5=Excellent)
Q.3	How would you rate the video quality? (1=Bad; 2=Poor; 3=Fair; 4=Good; 5=Excellent)
Q.4	How would you rate the responsiveness of the system? (1=Bad; 2=Poor; 3=Fair; 4=Good; 5=Excellent)
Q.5	If there is any problem with the system's responsiveness, what is the main component?
Q.6	How would you rate your comfort? (1=Bad; 2=Poor; 3=Fair; 4=Good; 5=Excellent)
Q.7	How would you rate the controllability of the system? (1=Bad; 2=Poor; 3=Fair; 4=Good; 5=Excellent)
Q.8	How would you rate your experience of interacting with the system in this test run? (1=Bad; 2=Poor; 3=Fair; 4=Good; 5=Excellent)

main effects and interaction terms remain valid for analysis using ANOVA. Post-hoc pairwise comparisons were conducted using Tukey's Honest Significant Difference (HSD) test to identify specific differences between conditions. Effect sizes were calculated using generalised eta squared to assess the magnitude of observed effects. This approach provided robust insights into how different system configurations influenced user performance.

Quantitative ratings

The performance analysis followed a mixed-method approach (see Table 5) inspired by the QoE domain, integrating objective measures, and qualitative feedback [48]. Quantitative analysis focused on perceived system responsiveness and controllability; these metrics are used to assess interactive system performance.

- System responsiveness refers to the perceived delay between user input and observable system reaction, particularly in visual feedback. This was assessed using participant ratings on how quickly and accurately the system responded to their control inputs.
- Controllability describes the user's perceived ability to reliably and accurately direct the system's actions, particularly regarding steering and navigation precision.*System's responsiveness and controllability:* These metrics were captured using recurring questionnaire responses (Table 6 - Q.4 and 7) on a 5-point Likert scale: Bad = 1, Poor = 2, Fair = 3, Good = 4, and Excellent = 5. The Likert scale's visual presentation is shown in Fig. 8. The mean opinion scores (MOS) were calculated from the participants' scale responses.

The normality assumption was evaluated using the Shapiro-Wilks test [72], with additional inspection via Q-Q plots [73]. Given the robustness of ANOVA to minor violations of normality, we proceeded with parametric analysis, employing a repeated-measures ANOVA to evaluate differences in Mean Opinion Scores (MOS) [74].

We applied Tukey's HSD as a post-hoc test to conduct pairwise comparisons following the repeated-measures ANOVA. Tukey HSD was selected for its suitability for all pairwise comparisons within an ANOVA framework, providing strong control over Type I error rates across multiple comparisons [75]. To quantify the magnitude of observed differences, we used Cohen's guidelines for interpreting eta-squared (η^2) to determine how much of the variance in the dependent variable can be explained by the independent variable [76].

According to Cohen's guidelines for eta-squared interpretation, this effect size is considered large ($\eta_g^2 > 0.14$), demonstrating a substantial influence of test configurations on users' performance [76]. Cohen's thresholds for interpreting effect sizes are defined as *small:* $\eta_g^2 = 0.01$, *medium:* $\eta_g^2 = 0.06$, *large:* $\eta_g^2 = 0.14$. In addition, Cohen's thresholds for effect sizes in terms of d are *small:* $d = 0.2$, *medium:* $d = 0.5$, and *large:* $d = 0.8$.

Qualitative data

As part of the mixed-methods approach, performance analysis included objective measures, quantitative ratings, and qualitative feedback. This section presents the analysis of users' open-ended questionnaires (e.g., Q.2, Q.3, and Q.6 in Table 7), which provided deeper insights into user perceptions of system responsiveness, usability, and multisensory impacts:

- **System Responsiveness:** Participants' feedback on system responsiveness provides deeper insights into system responsiveness by uncovering specific reasons behind users' ratings.
- **System Usability:** Insights into ease of use and interface intuitiveness derived from open-ended questions.
- **Multisensory Impacts:** Feedback on how sensory elements, such as auditory cues (e.g., truck sound), influenced task performance.

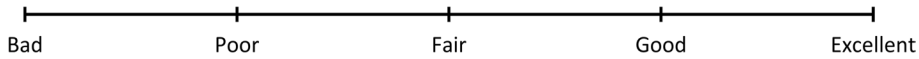


Fig. 8 Likert-scale visual presentation in questionnaires

Table 7 Questions formulated for the open-ended questionnaire

Q.No	Questions asked from participants after each trial in open-ended questionnaire
Q.1	What is your general experience of working with the Remote platform?
Q.2	Did the sound of the truck help you in driving?
Q.2 - a	Regarding the previous question if your answer is "yes", please explain how it was helpful
Q.3	Did you find that the system, at some point, became practically impossible to perform or work with?
Q.3 - a	Regarding the previous question, if your answer is "yes", please add what the main problem was
Q.4	How would you rate the impact of different video quality on depth estimation?
Q.4 - a	How do these changes affect your depth perception?
Q.5	How would you rate the impact of different Fields of view on depth estimation?
Q.5 - a	How do these changes affect your depth perception?
Q.6	What improvements to this system would further support these types of tasks (as tried in the tests)?
Q.7	What other applications can you imagine would benefit from these kinds of systems?

The approach is largely inspired by the UX domain. Thematic analysis was employed to categorise responses, revealing patterns in user perceptions and suggestions. For the recurring and open-ended questionnaire data, we employed Grounded Theory to inductively derive themes and insights. Following a bottom-up approach, we began with raw data, systematically coding participant responses to identify recurring patterns. This iterative process enabled the formation of categories and themes that represent core aspects of the user experience, particularly responsiveness, multisensory feedback, and system controllability under different configurations. This method was essential for uncovering user-driven insights that may not have been anticipated with predefined UX metrics, making it well-suited for investigating the unique challenges of remote control operation in a simulated yet realistic setting.

Data analysis - depth perception (RQ2)

To address RQ2, measurements were categorised into objective data, user ratings, and users' qualitative reflections (shown in Table 5). Objective data focused on the Average Depth Estimation Deviation (ADED), which quantified the accuracy of participants' depth judgments. Quantitative ratings assessed the impacts of Field of View and Video Quality on depth perception, as well as participants' confidence in their depth estimates, using predefined scales. Qualitative feedback provided insights into the experiential aspects of FoV and Video quality on depth perception, capturing user perspectives that may not be addressed by quantitative metrics alone (Table 5).

Objective data

Average Depth Estimation Deviation (ADED): The ADED quantified the accuracy of participants' depth judgments. In the depth perception task, participants estimated the distance to defined red and yellow boxes (see Fig. 4 - right) using a reference scale marked as "X" (Fig. 4 - right) and reported their estimations with one decimal precision (e.g., 1.2 X).

The difference between the estimated depth and the actual distance is denoted as $a_{i,j,k}$ for participant i, trial j, and object k. The ADED metric $\mathcal{Z}j,k$ for trial j with N participants is defined as:

$$\mathcal{Z}j,k = \frac{\sum_{i=1}^{N} a_{i,j,k}}{N} \tag{4}$$

Quantitative ratings

Users rated their experience on a 5-point Likert-like scale, assessing the effects of FOV and VQ on depth perception (Table 7 - Q.4 and Q.5). Participants also rated their confidence in estimating the depth to defined targets (Table 6 - Q.7) while interacting with the teleoperation across various visual interface configurations.

Qualitative data

As presented in Table 5, "Depth Perception" was assessed using two criteria: the effect of FoV on depth perception and VQ on depth perception. The influence of FoV and VQ corresponds to responses from Q.4-a and Q.5-a, measured in the open-ended questionnaire (Table 7). A bottom-up approach was employed to analyse the qualitative data, systematically coding responses to identify recurring patterns and derive insights.

Data analysis - General User Experience (RQ3)

General UX measurements were categorised into quantitative ratings and qualitative feedback (shown in Table 5). Quantitative ratings included users' comfort, the Interaction Experience Rating (IER), and simulator sickness. Qualitative feedback, captured through the General Remote Experience (GRE) and system improvement measures, provided insights into users' overall impressions of the system and suggestions for enhancing usability and functionality.

Quantitative data

IER, user comfort, and Simulator sickness comprised the UX metrics in this study. User comfort and IER were assessed using Likert-like scales in response to recurring questions (Q.6 and Q.8) outlined in Table 6. In this study, we distinguish between 'comfort' and 'IER' to capture different dimensions of user experience. Comfort focuses on the physical and cognitive ease of system use, while IER reflects users' holistic evaluation of the interaction, including intuitiveness, enjoyment, and perceived system quality. Simulator sickness was evaluated using the SSQ, completed once before and once after the test runs to capture potential changes in symptoms. The SSQ, adapted from Kennedy et al. [66, 77], includes 16 symptoms rated on a 4-point scale (0 = "Not at all" to 3 = "Severe"). Scores were aggregated into subscales for Nausea, Oculomotor, Disorientation, and Total Score, following the defined formula and weightings. Pre- and post-experiment scores were analysed using paired sample t-tests to assess changes in sickness levels across sessions.

Qualitative data

Qualitative data were analysed using an inductive, data-driven approach informed by grounded theory and structured through Braun and Clarke's thematic analysis framework. First, all open-ended responses from UX-related questions (Q.1, Q.6, Q.7 as outlined in Table 7) were reviewed by the first author to ensure familiarity with the content. Following grounded theory principles, recurring concepts and expressions were initially coded without a predefined framework, allowing themes to emerge directly from the data. These emergent codes were then iteratively compared, grouped, and refined into higher-level categories.

These categories were subsequently structured using Braun and Clarke's six-phase thematic analysis method [78], ensuring a systematic and transparent approach to theme development. As patterns stabilised, the codes were organised into an overarching experiential dimension: hedonic and eudaimonic aspects. These thematic categories were informed by existing UX literature but grounded in the participants' own language. The coding was conducted using Dovetail, a qualitative analysis platform that supported traceability and organisation of quotes. These categories were then structured using Braun and Clarke's six-phase thematic analysis method.

- Hedonic aspects reflect users' reported enjoyment, pleasure, and satisfaction during the teleoperation tasks.
- Eudaimonic aspects capture feedback on meaningful engagement, perceived challenge, and moments of learning or adaptation.

While eudaimonic aspects were initially considered during coding, the final analysis focused on hedonic experiences, as the study design and single-session nature of the experiment did not allow for meaningful assessment of long-term eudaimonic factors. The analysis was conducted by the first author, and no second coder was involved due to practical constraints. The codes were revisited multiple times to ensure internal consistency and interpretive clarity. Representative participant quotes were selected to illustrate themes and retain the user voice. The resulting themes provided a richer interpretation of participants' experience and interactions with the system beyond what was captured in quantitative measures.

Results

This section presents the results of the experiment, derived from the analysis methods primarily described in Sect. "Method"

RQ1: Performance

Performance - objective data

Figure 9 illustrates the truck deviation from the main track across different test conditions (trials), serving as a performance metric. A factorial ART ANOVA was conducted to assess the effects of Latency, Field of View, and Video Quality on deviation. The results indicated a statistically significant effect of test conditions on performance, F(4.1, 77.9) = 4.815, p = 0.001, $\eta_g^2 = 0.196$, indicating a large effect size.

Post-hoc pairwise comparisons using Tukey HSD tests revealed that T18 (LoQual HiLat NarrowFOV) resulted in significantly higher deviations, indicating lower performance, compared to several other conditions. In particular, T18 had a significantly greater deviation than T1 (HiQual LoLat WideFOV) (diff = 3.16, $p < 0.001$), and T10 (HiQual LoLat NarrowFOV) (diff = 2.74, $p < 0.001$).

These findings highlight that higher latency, lower video quality, and a narrower field of view together lead to the greatest degradation in performance, as evidenced by the highest deviation in T18 (LoQual HiLat NarrowFOV). Conversely, T1 (HiQual LoLat WideFOV) demonstrated the lowest deviation, suggesting optimal user performance under high video quality, low latency, and wide field of view conditions.

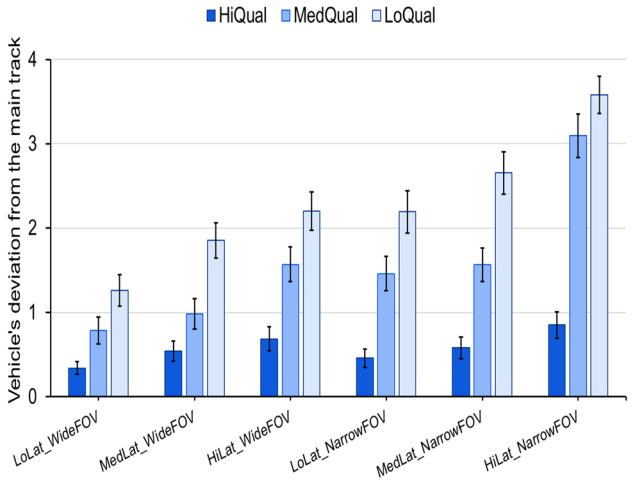


Fig. 9 Truck deviation from the main track serves as a performance metric, with greater deviation indicating lower performance. Error bars show 95% confidence intervals, representing variability across conditions

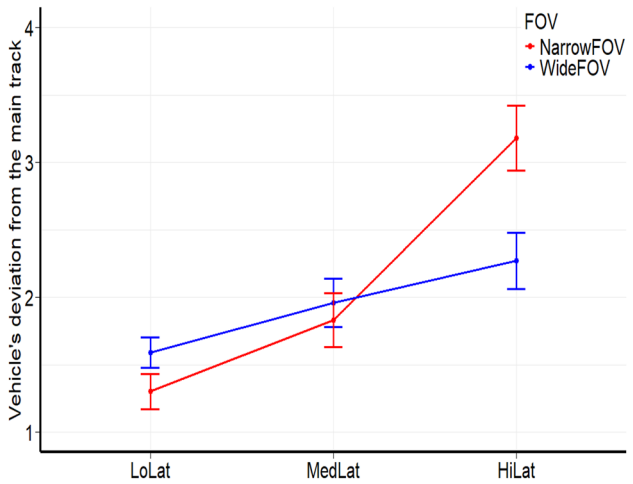


Fig. 10 Interaction effects between FoVs and Latency on the mean absolute difference (Truck deviation from the defined track) in user performance metric. Greater deviation indicates lower performance scores. Error bars indicate 95% confidence intervals, reflecting variability across conditions

The interaction plot in Fig. 10 illustrates the relationship between latency levels (LoLat, MedLat, and HiLat) and Field of View (FoV) conditions (WideFoV and NarrowFoV) on performance, as measured by the mean absolute difference in truck deviation from the defined track. Error bars represent 95% confidence intervals, reflecting variability across conditions. A higher mean absolute difference indicates lower performance.

The interaction effect between Latency and FoV was analysed using ART ANOVA, yielding a statistically significant result ($F(4, 429) = 13.33$, $p < 0.001$). According to Cohen's guidelines, the calculated eta squared ($\eta_g^2 = 0.111$ suggests a medium effect size, indicating that the combined influence of latency and FoV has a meaningful impact on the truck's deviation. However, it does not reach the threshold of a full large effect. At lower latency levels (LoLat and MedLat), both WideFoV and NarrowFoV conditions perform similarly, with NarrowFoV exhibiting slightly lower mean absolute differences, suggesting marginally better performance. However, as latency increases to HiLat, the performance gap between FoV conditions widens considerably. The NarrowFoV condition exhibits a significant increase in mean absolute difference, indicating a greater decline in performance under high latency compared to WideFoV. This trend suggests that while both FoV conditions are affected by increasing latency, NarrowFoV is more sensitive to high latency, leading to a larger reduction in performance.

These findings emphasise the importance of considering both latency and camera FoV in teleoperation settings, as their interaction substantially influences user performance and control precision.

Performance - user ratings

System responsiveness The results for the responsiveness of the system, measured by MOS indicates statistically significant differences based on the different conditions applied. The ANOVA analysis yielded a significant effect, with $F(6.2, 117.73) = 6.642$, $p < 0.001$, and $\eta_g^2 = 0.138$ suggesting a medium effect size.

Based on Tukey's HSD test for multiple comparisons, several test conditions show significant differences in system responsiveness (Fig. 11), particularly involving comparisons with T 1 (HiQual Lowlat WideFoV) and T 2 (MedQual Lowlat WideFoV), highlighting these conditions as critical for users' responsiveness.

- T 1 (HiQual LoLat WideFoV) vs. T 16 (HiQual HiLat NarrowFoV): This comparison yielded a strong effect, with a Cohen's *d* of 1.64, a mean difference of 1.35, and an adjusted *p*-value of 0.006. This indicates a substantial reduction in perceived responsiveness at *T 16* compared to *T 1*.
- T 1 (HiQual LoLat WideFoV) vs. T 9 (LoQual HiLat WideFoV): A strong effect size was observed with a Cohen's *d* of 1.49, a mean difference of 1.30, and an adjusted *p*-value of 0.01, indicating a notable drop in responsiveness.
- T 1 (HiQual LoLat WideFoV) vs. T 7 (HiQual HiLat WideFoV): This comparison also showed a strong effect with a Cohen's *d* of 1.45, a mean difference of 1.25 and an adjusted *p*-value of 0.018, emphasising a significant responsiveness decrease at *T 7*.

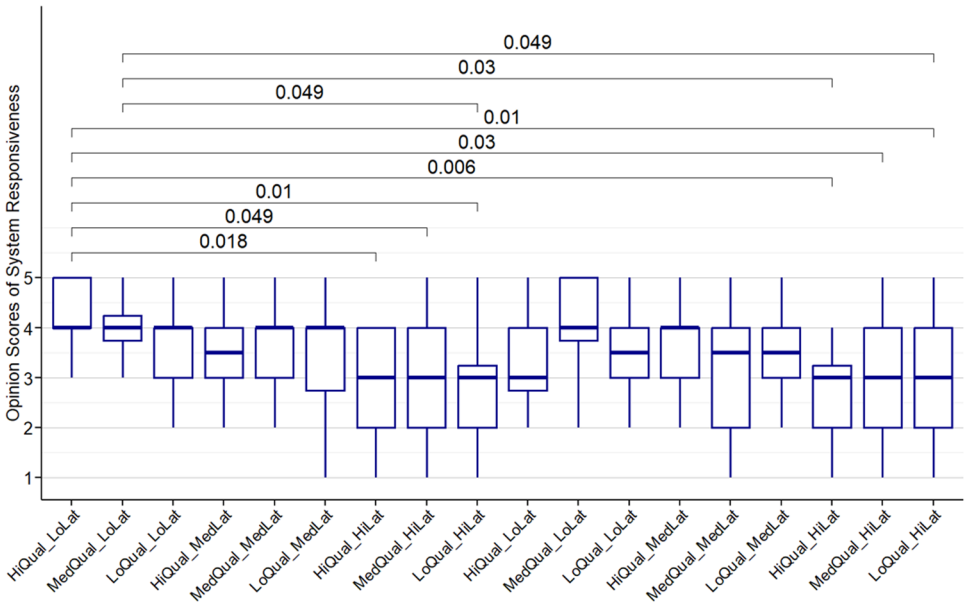


Fig. 11 Boxplot of system responsiveness (MOS) grouped by FoV. The first nine conditions correspond to Wide FoV, the last nine to Narrow FoV. Boxes show interquartile range with medians; whiskers indicate data within 1.5× IQR, and outliers are marked. Significant pairwise differences based on Tukey post-hoc tests are annotated with p-values

- T 2 (MedQual LoLat WideFoV) vs. T 16 (HiQual HiLat NarrowFoV): This comparison also indicates a strong effect, with a Cohen's *d* of 1.42, a mean difference of 1.20, and an adjusted *p*-value of 0.03, reflecting a substantial difference in responsiveness.
- T 1 (HiQual LoLat WideFoV) vs. T 17 (MedQual HiLat NarrowFoV): Here, Cohen's *d* is 1.24, with a mean difference of 1.20 and an adjusted *p*-value of 0.03, indicating a considerable reduction in responsiveness.

In summary, the comparisons involving *T1: HiQual LoLat WideFoV* and *T16: HiQual HiLat NarrowFoV* exhibited the largest effect sizes, underscoring these test conditions as critical periods for degradation in perceived responsiveness. The consistent pattern of significant differences and large Cohen's *d* values, particularly in comparisons with *T16*, demonstrates the adverse impact of these conditions on system responsiveness.

The boxplot in Fig. 11 shows that higher latency conditions (HiQual-HiLat) generally resulted in lower MOS ratings for system responsiveness, especially in the Narrow-FoV configurations, compared to the Low Latency conditions. Furthermore, configurations with High Video Quality tended to receive better MOS scores than those with Medium or Low video quality, particularly under lower latency.

The impact of FoV, latency, and video quality on responsiveness is evident, as lower video quality and higher latency led to diminished user-perceived responsiveness, highlighting the importance of these parameters in teleoperation performance.

System controllability

The results for the system's controllability, measured by MOS, indicate no statistically significant differences across the various conditions tested. The ANOVA results showed

$$F(6.06, 115.19) = 1.246, p = 0.288, \eta_g^2 = 0.03.$$

Performance - qualitative reflections

System responsiveness The results shown in Fig. 12 highlight the distribution of user feedback regarding the factors affecting system responsiveness; 62% of participants stated a component negatively impacts the system's responsiveness. Among these responses, 33% (119 out of 360 occasions) attributed responsiveness issues to the "Game controller", indicating this component is the most significant factor impacting the system's responsiveness. Additionally, 15% of responses (55 out of 360) reported the "Provided view" as a negative factor affecting responsiveness. A smaller percentage, 8%, identified both the game controller and the provided view as sources of responsiveness issues. Finally, latency, as represented in the "Other" category, was mentioned by approximately 5% as a contributing factor to reduced system responsiveness. These findings suggest that the game controller is the primary area needing enhanced responsiveness improvement.

System usability

In the subjective analysis of system usability, nine out of twenty participants reported experiencing moments when

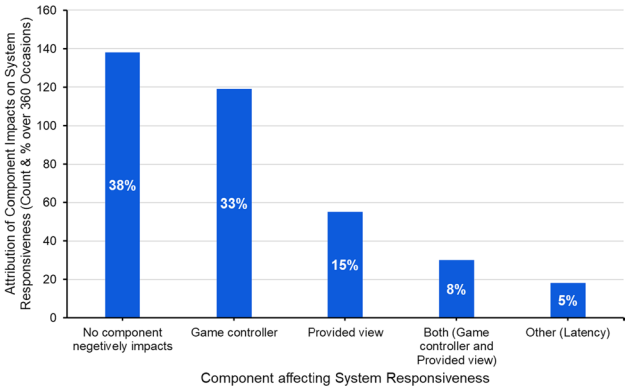


Fig. 12 Users' feedback on components affecting system responsiveness, based on 360 responses (20 users across 18 trials). Components include the game controller, provided view, latency, and their combinations

the system became practically unusable. The feedback primarily emphasised latency and control responsiveness issues, directly impacting usability.

Several users indicated that high latency made the system challenging to control, as there was a noticeable latency between commands and the truck's response in some trials (e.g., "It was very difficult to drive when the latency was high," and "In the last test (refer to T16), I felt it was uncontrollable"). Additionally, participants described difficulties in controlling movement precision, with some reporting that slight controller inputs led to unexpectedly large or no movements, creating inconsistency in the system's responsiveness (e.g., "Movements' steps were not easily controllable. Sometimes a quick push resulted in a large movement, and sometimes no movement at all."). It could be attributed to either Glass-to-Glass latency or game controller latency.

Video streaming issues further compounded usability challenges, with some participants experiencing latency that affected their ability to accurately anticipate obstacles. One user noted almost crashing due to the delayed video response ("The video latency made me almost crash into the red box because of the late video response"). This feedback highlights significant areas for improvement in system responsiveness, particularly under high-latency conditions, to ensure reliable usability.

Multisensory feedback

In the qualitative analysis of multisensory effects on driving performance, 15 out of 20 participants reported that the sound of the truck had a positive impact on their driving experience. Many responses indicated that the audio feedback supported users' control and timing, particularly in scenarios where visual feedback alone was insufficient or delayed.

Participants highlighted several ways in which the truck's sound improved their performance. Some noted that it helped confirm the truck's movement and enabled them to estimate the latencies between the control input and the vehicle's response. This awareness allowed users to compensate for latency by reacting sooner and estimating streaming delay based on sound cues ("I could estimate the live streaming delay by considering the time I pressed the gamepad button, the time I heard the truck moving sound, and the time the truck moved on the screen. This estimation helped me to react sooner than the time that actually needed based on screen", "I could see the delay, either in the control or in the visual feedback, but having also audio feedback helps estimate the amount of the delay and act accordingly when "driving"). Others emphasised the reassurance provided by the sound, as it confirmed that their control input was acknowledged by the system ("The sound of the truck assures you that your control input is being quickly fed to the truck and you won't need to give more elongated presses on the controller stick for the truck to move.").

Several participants also observed that audio cues supported their ability to judge the truck's speed, acceleration, or general responsiveness, which enhanced the intuitiveness of the driving experience ("I could sense the speed or if the truck was driving", "Through the change of the sound, I can get informed if the truck is accelerating"). In cases of low video quality or screen latency, users found that audio feedback provided essential information on the truck's status and responsiveness ("It was easier to determine and compensate for latency", "Even if the video quality was awful, I could know when the car started moving"). These results underscore the beneficial role of multisensory feedback, with sound serving as a compensatory cue for latency and visual limitations, ultimately improving users' sense of control and system responsiveness during remote driving.

RQ 2: Depth perception

Depth perception - objective Data

ADED values for object 1 and object 2 remain relatively consistent across conditions, with object 2 (ADED-object2: 77.6 cm) generally showing slightly higher deviations than object 1 (ADED-object1: 68.6 cm), indicating that depth estimation for object 2 was slightly more challenging. However, this difference was not statistically significant according to the ANOVA results, and thus, no definitive conclusions can be drawn.

Depth perception - user ratings

Effects of FoV and VQ on depth perception The violin plot in Fig. 13 shows the distribution of Opinion Scores for the impacts of FoV and VQ on depth perception. The mean, represented by the red (FoV) and blue (VQ) circles, suggests that FoV had a slightly higher perceived impact on depth perception than VQ. Both distributions follow a similar trend, but the VQ scores exhibit greater spread, indicating more variability in participants' ratings. This suggests that participants had more inconsistent perceptions of how VQ affected depth perception, while FoV elicited more consistent responses.

The paired t-test revealed a statistically significant difference between "FoV on depth perception" (M = 3.6) and "VQ on depth perception" (M = 2.6), $t(19) = 2.94$, $p = 0.008$, effect size ($d=0.66$) indicates a medium to large effect size, according to Cohen's guidelines, indicating that the perceived impact of FoV on depth perception was consistently higher than the perceived impact of VQ on depth perception. This suggests that FoV plays a more critical role in depth perception than VQ in this experimental setup.

Users' confidence in depth estimation

According to the ANOVA test, no significant differences were observed across conditions. Confidence ratings and depth estimation values followed similar trends across all conditions. No significant differences were observed in these measures.

Depth perception - qualitative reflections

To determine which factor, FoV, VQ, or Latency, had the most significant impact on depth perception based on user feedback, we compared the proportion of participants reporting the effects for each factor.

Impact of FoV on depth perception

Thirteen out of twenty participants highlighted the significant role of FoV in their depth estimation. Users' insights are categorised as follows:

- *Perceived Limitations of NarrowFoV:* Four participants noted that the NarrowFoV reduced visible context, making depth estimation more difficult. One participant commented that the reduced field of view made it harder to assess the truck's position relative to waypoints, while another noted that it limited their ability to see enough of the path ahead. Example of the users' statements: *"The narrower views made me rethink my depth estimates*, *"If the field of view is narrowed down, the amount of control inputs to left and right will be greatly affected in a negative wa."*, *"As getting narrower, I estimate the distance less and less well"*, *"Narrower FoV affects the depth perception in the sense that it is more*

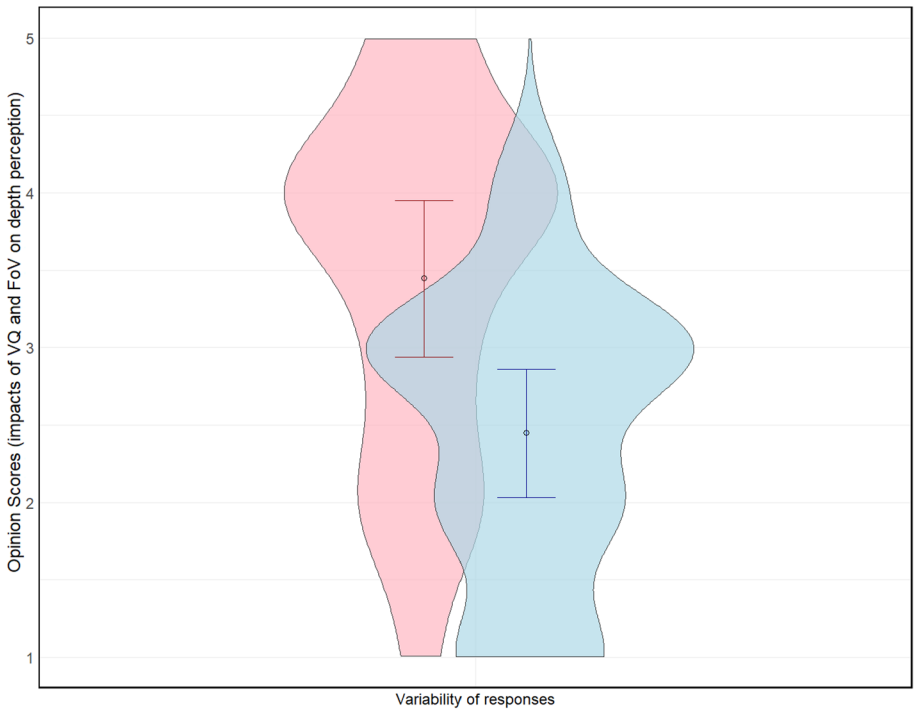


Fig. 13 Violin plots showing the distribution of Opinion Scores for the perceived impacts of FoV and VQ on depth perception. Wider sections of the plot represent a higher response density. The overlaid points represent the mean, with confidence intervals indicated by error bars. The confidence intervals for FoV are defined in red, and VQ is defined in blue

difficult to understand if you have actually reached the waypoints."

- *Perceptual distortion due to wide FoV:* Two participants referred to perceptual distortion associated with a wider FoV: *"These wider changes had a big impact as the reference sometimes felt closer or further away from the point of view"*. This describes inconsistencies in perceived depth resulting from the distortion of spatial relationships under different viewing angles. It implies the user experienced instability in depth estimation caused by changes in FoV, which is a common challenge in wide-angle teleoperation interfaces. *"Having higher FOV makes the fish eye effect, so the distances on the screen are not realistic. The farther from the centre of the image, the greater the expansion of the distances"*. This is a direct reference to the fisheye distortion, indicating that peripheral expansion misrepresents actual distances.
- *FoV Aligning with Human Vision:* One participant emphasised that WideFoV provided better spatial context, aligning with human vision. *"A more horizontal view makes it more similar to human vision."*
- *Perceptual Variability with Changing FoV:* Seven participants reported that depth estimation was inconsistent across different FoV configurations. They noted variations in object sizes and difficulty in accurately judging distances, highlighting the influence of visual changes on perception. Example of users' comments: *"With different FoV, objects are looked at with different sizes, which influences the depth estimation."*, *"Field of views changes have an important effect on depth perception"*, *"Field of view seemed to affect depth perception, but I noticed that once."*, *"It impacts the distance I perceive with the objects."*, *"Changes in FoV, Broaden or narrow my view."*
- *No Noticeable Effect of FoV:* Six participants reported no significant impact of FoV changes on their depth perception. They described the variations as negligible or not influencing their estimation tasks. Example of Users' statements: *"I did not have a feeling of the difference with the changes of the field of view."*, *"Almost nothing."*, etc.

Impact of VQ on depth perception

Participants' qualitative feedback regarding the impact of VQ on depth perception is categorised into distinct themes, highlighting both the strengths and challenges of varying VQ levels.

- *Reflection VQ Impacts on Depth Perception:* Three participants noted that lower video quality introduced blurriness, making depth estimation more difficult and causing visual strain. They emphasised the importance of maintaining a baseline resolution to ensure the clarity of critical visual elements. Two participants emphasised that higher video quality enhanced depth perception. They found that improved resolution and clarity increased their confidence and precision in estimating distances. Five participants noted that video quality influenced their depth perception, but did not specify whether high or low quality was more impactful. Examples of users' statements: *"The reference line got blurry, which made it hard to check if the task was accomplished."*, *"Changes (in VQ) may impact my action on the controller."*, "Better resolution gave a sense of better depth perception."
- *"Sometimes more distant objects felt closer, although the reference almost always helped."*, etc.
- *Impact of FoV vs VQ on Depth Perception:* One participant reported that changes in aspect ratio or scaling had a more negative impact on depth perception than video quality. *"The aspect ratio and lens perspective had a much higher impact on depth estimation when compared to the resolution during the experiments."*
- *No Noticeable Effect of VQ:* Seven participants reported that changes in video quality had little to no effect on their depth perception. They found the provided resolution sufficient for completing tasks effectively. These participants indicated that video clarity was less critical for their depth estimation. Examples of users' comments:*"It didn't really [affect it], the lowest quality was good enough for the edges of the boxes to be somewhat sharp."*, *"With the current video quality, I was also able to move the truck with ease, hence improvement in video quality might not be as significant a jump."*, *"It (VQ) didn't really affect my depth perception."*, etc.

Impact of latency on depth perception

Fifteen out of twenty participants found that latency has minimal or no effect on depth perception. While most participants reported minimal direct effects of latency on depth perception, their feedback revealed indirect impacts through related challenges, such as task performance, cognitive strain, and user adaptation. The insights are categorised into the following themes:

- *Minimal Direct Effect on Depth Perception*: Five participants suggest that any observed challenges were predominantly related to task performance and control precision rather than depth perception itself.

- *Impact on Performance*: item Latency was identified as a factor affecting the ability to smoothly control and navigate the truck. Participants expressed frustration with delayed responses, which hindered their precision. *"The latency in controlling the system caused a disturbance in taking the turn required."*, *"Depth perception, if degraded, would make it difficult to move the truck."*
- *Adaptation Over Time*: One participant noted that with practice, they were able to adjust to the latency and compensate for its effects. *"With latency, I find that I need a warm-up period until I get used to it and can drive accurately with the delay."*
- *Cognitive and Physical Strain*: Two participants highlighted the mental and physical challenges caused by latency, which negatively affected their confidence and overall performance. *"It makes the user feel tired and low-confident for driving."*, *"I lost my perception and accuracy."*

RQ 3: General user experience

General UX - user ratings

Users' comfort A repeated-measures ANOVA revealed a statistically significant effect of video degradations, including VQ, latency, and FoV, on user comfort ratings, $F(6.4, 121.65) = 5.256, p < 0.001, \eta_g^2 = 0.082$. The generalised eta-squared effect size ($\eta_g^2 = 0.082$) indicates a medium impact of these degradations on user comfort during the teleoperation task.

Post-hoc comparisons using the Holm correction showed the following key differences:

- User comfort ratings for **T1** (*HiQual_LoLat_WideFoV*) were significantly higher than those for **T17** (*MedQual_HiLat_NarrowFoV*), with a mean difference of $\text{statistic} = 4.70, p_{\text{adj}} = 0.024$.
- Comfort ratings for **T1** (*HiQual_LoLat_WideFoV*) were also significantly higher than those for **T18** (*LoQual_HiLat_NarrowFoV*), with a mean difference of $\text{statistic} = 4.82, p_{\text{adj}} = 0.018$.
- Ratings for **T3** (*LoQual_LoLat_WideFoV*) were significantly higher than for **T18** (*LoQual_HiLat_NarrowFoV*), with a mean difference of $\text{statistic} = 5.60, p_{\text{adj}} = 0.003$.
- Finally, comfort ratings for **T4** (*HiQual_MedLat_WideFoV*) were significantly higher than those for **T18** (*LoQual_HiLat_NarrowFoV*), with a mean difference of $\text{statistic} = 4.68, p_{\text{adj}} = 0.025$.

The findings indicate that video degradations–such as low video quality, high latency, and narrow field of view (e.g., **T18**)–significantly reduced user comfort compared to configurations with high video quality, low latency, and a wide field of view (e.g., **T1**).

Simulator sickness

The SSQ was administered before and after the experiment, assessing Disorientation, Nausea, Oculomotor, and Total Score. The post-test Total Score (91.1) was mainly driven by elevated Oculomotor (11.0) and Disorientation (7.66) scores, while Nausea remained low (5.72). Overall, simulator sickness was mild and unlikely to have affected task performance or system usability.

Interaction experience rating (IER)

A repeated-measures ANOVA was conducted to assess the effect of different visual configurations on users' general experience scores while interacting with the LRCP. The statistical analysis revealed a significant effect, as indicated by the following results: F(6.9, 160.7) = 5.8, p < 0.001, $\eta_g^2 = 0.09$ (shown in Fig. 14).

This indicates that the IER scores varied significantly across different trials, with effect size ($d = 0.62$) representing a medium effect, suggesting a moderate impact of visual configurations on overall user experience scores.

A Tukey HSD post-hoc test was conducted to further examine the differences between the different applied configurations. The test results showed the following significant pairwise comparisons:

- The IER score at T18 (LoQual HiLat NarrowFoV) was significantly lower than at T1 (HiQual LoLat WideFoV), with a mean difference of diff = −1.08, p = 0.03.
- The IER score at T2 (MedQual LoLat WideFoV) was significantly higher than at T18 (LoQual HiLat NarrowFoV), with a mean difference of diff = 1.04, p = 0.04.

These results suggest that IER scores at T18 (LoQual HiLat NarrowFOV) were significantly lower than those at T1 (HiQual LoLat WideFoV), and IER scores at T2 (MedQual LoLat WideFoV) were significantly higher than those at T18. No other significant differences were found between the remaining test configurations.

Impact of video quality

The analysis revealed a statistically significant effect of VQ and latency configurations on IER scores, F(6.9, 132.3) = 6.9, p < 0.001, $\eta_g^2 = 0.16$. This indicates that the tested

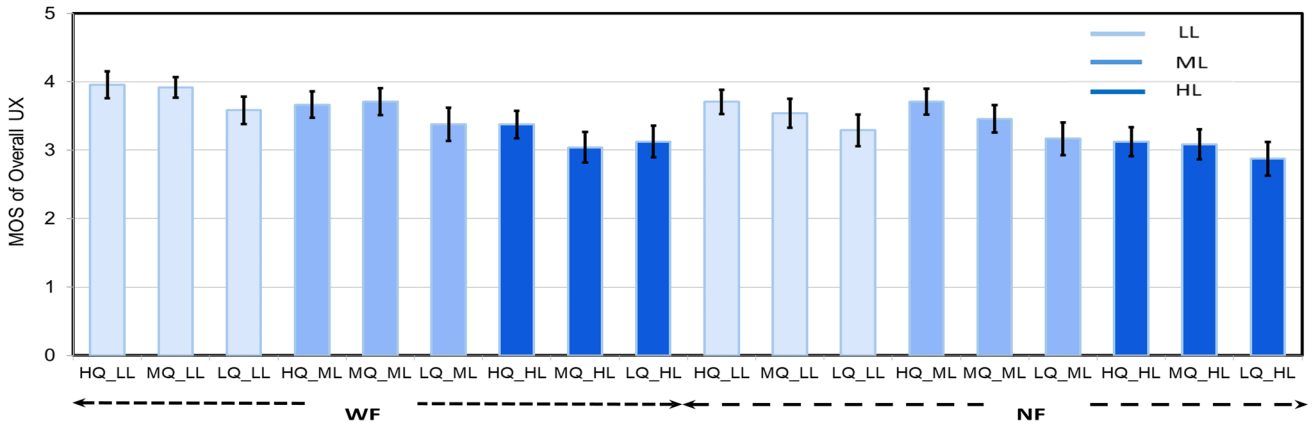


Fig. 14 MOS of Interaction Experience Rating (IER) across different test conditions (trials). Error bars represent 95% confidence intervals, showing variability in user ratings

configurations significantly influenced how participants rated their experience, with a medium to large effect size ($\eta_g^2 = 0.16$) highlighting the importance of video quality and latency in shaping UX.

Post-hoc pairwise comparisons, corrected using the Holm method, revealed the following key differences between conditions:

- The IER score at **T18** (*LoQual_HiLat_NarrowFoV*) was significantly lower than at **T1** (*HiQual_LoLat_WideFoV*), with a mean difference of diff = −1.08, p = 0.03.
- The IER score at **T18** (*LoQual_HiLat_NarrowFoV*) was also significantly lower than at **T12** (*LoQual_LoLat_WideFoV*), with a mean difference of diff = −0.96, p = 0.03.
- A smaller but still significant difference was observed between **T1** (*HiQual_LoLat_WideFoV*) and **T15** (*LoQual_MedLat_NarrowFoV*), with a mean difference of diff = −0.75, p = 0.05.

These findings indicate that conditions combining low video quality, high latency, and a narrow field of view consistently yielded lower IER ratings than configurations with high video quality, low latency, and a wide field of view.

General user experience - qualitative reflections

System improvement

Participants provided various suggestions to enhance the system's performance and usability. The most frequently mentioned improvement, noted by 7 out of 20 participants, was reducing latency, including minimising delays between the remote controller and vehicle actions. Enhancing video quality and providing additional camera views were also significant, with 6 participants highlighting the need for wider fields of view or multiple perspectives (e.g., ceiling-mounted or lateral views). A few participants (3 out of 20) suggested realism-enhancing features, such as a steering wheel-like controller and head-turning functions, to better align the system with real-world driving conditions. Other notable suggestions included faster processing speeds (2 participants), adjustable speed controls (1 participant), and the potential use of virtual/augmented reality to improve immersion (1 participant).

These users' feedback indicates a clear preference for latency, camera angles, and video quality improvements, with some users advocating for AR and VR integration to further support task performance. Addressing these aspects could potentially enhance system usability and provide users with an experience closer to real-world operation.

General remote experience

The general user experience with the remote-control system was analysed through *Hedonic aspects*. Participants highlighted several positive and enjoyable elements:

- *Perceived Enjoyment:* Under low-latency and high-quality video conditions, users described the experience as smooth and enjoyable. For instance, *"It was a really nice experience"*, suggests high hedonic value when system responsiveness was optimal.
- *Perceived Novelty and Engagement:* Participants who were new to remote control noted the uniqueness of the task, *e.g., "Fun, interesting experience"*, highlighting how unfamiliar configurations (e.g., narrow FOV or video degradation) affected engagement levels.
- *Perceived Ease of Use:* In conditions with wider FOVs and low delay, some users reported better control and flow: *"Very interesting to control from a remote*

distance", indicating how system configurations shaped their sense of control and satisfaction.

Discussion

This study investigated the impact of Video Quality, Field of View, and latency on users' performance, depth perception, and overall user experience using a lab-based remote control system. By adopting a user-centred approach [79] and integrating mixed-method approaches from QoE and UX domains [58], the research addressed the challenges of 3D understanding through 2D visual interfaces in teleoperation, a critical issue in the construction industry [17, 18]. The combination of quantitative and qualitative data offered a comprehensive perspective on users' visual perception and interaction under constrained visual conditions.

RQ 1: Impact of VQ, FoV, and Latency on Users' Performance

To answer RQ1, performance was evaluated through objective, quantitative, and qualitative measures. Objective data focused on the truck's deviation from the defined track as the primary performance metric. Quantitative data from the QoE domain assessed system responsiveness and controllability, while qualitative feedback from the UX domain explored participants' perceptions of system responsiveness, usability, and multisensory impacts.

The objective results demonstrate a clear interaction between latency and FoV conditions, showing that performance is influenced by their combined effects. At lower latency levels (LoLat and MedLat), performance across WideFoV and NarrowFoV is similar, with NarrowFoV slightly outperforming WideFoV due to its potential to focus users' attention on critical visual cues. However, as latency increases to HiLat, a significant performance gap emerges. Performance significantly declined under the NarrowFoV condition, as indicated by an increased mean absolute difference, whereas the WideFoV condition maintained relatively stable performance. This suggests that NarrowFoV is particularly sensitive to high latency, possibly due to its limited spatial context, which makes it harder for users to compensate for delayed system responsiveness. Conversely, WideFoV provides a broader visual context, enabling users to better anticipate and adapt to latency-related disruptions. These findings underscore the critical interplay between FoV and latency in teleoperation design, emphasising the need for latency-optimised systems when using NarrowFoV and the robustness of WideFoV in high-latency environments.

System responsiveness, as assessed through quantitative data, showed that the comparisons between T1 (HiQual LoLat WideFoV) and T16 (HiQual HiLat NarrowFoV) revealed the largest effect sizes, identifying these conditions as critical for perceived responsiveness degradation. The significant differences and consistently large Cohen's d values, particularly in comparisons involving T16, underscore the detrimental impact of high latency combined with a narrow field of view on system responsiveness. Participants rated responsiveness lower under high-latency conditions, emphasising delays as a major factor disrupting their ability to make precise navigational adjustments. In contrast, controllability did not exhibit significant differences across conditions, suggesting that while responsiveness is highly sensitive to latency and visual configurations, overall system controllability remains unaffected.

Qualitative feedback provided valuable context to the quantitative findings, highlighting users' experiences and perceptions [80]. Participants frequently noted that delayed system responses made it challenging to predict and anticipate the truck's movements. In discussions of system responsiveness, participants often attributed the perceived latency to the game controller. However, technical latency measurements across the latency pipeline revealed that the game controller had low latency, and the delays were primarily caused by visual feedback. This misattribution underscores the importance of clear system communication to align user perceptions with technical realities. Incorporating intuitive elements like haptic feedback could help users better understand and respond to latency, enhancing both trust and usability [81]. The analysis highlighted key usability challenges, with latency and control responsiveness emerging as critical factors affecting system performance. High latency significantly disrupted control precision, making it difficult for users to navigate the truck accurately and reliably. Participants reported inconsistencies in system behaviour, where slight controller inputs occasionally led to exaggerated or unresponsive movements. These issues, likely linked to Glass-to-Glass latency or control system delays, were further exacerbated by video streaming delays that hindered users' ability to effectively anticipate obstacles. Such limitations emphasise the need to optimise system responsiveness and refine control mechanisms to enhance usability and user confidence. On the other hand, multisensory feedback, particularly audio cues, proved invaluable in mitigating some of these challenges. Participants frequently described how the truck's sound served as a real-time feedback mechanism, enabling them to estimate latencies and adjust their actions accordingly. Audio feedback reassured users that their inputs were registered, bolstering confidence in the system and reducing cognitive strain caused by visual feedback delays [82]. Additionally, sound cues helped users gauge speed and acceleration, which was particularly beneficial in low video quality or delayed streaming conditions.

These findings underscore the potential of integrating multisensory feedback, such as audio and haptic cues, to address usability issues, enhance system intuitiveness, and support a fluent remote driving experience.

RQ 2: Effect of VQ and FoV on Depth Perception

To answer RQ2, depth perception was analysed through objective, quantitative, and qualitative measures. Objective data focused on the Average Depth Estimation Deviation (ADED). Quantitative data from the QoE domain captured participants' ratings on the impact of VQ and FoV on depth perception and their confidence in depth estimation. Qualitative insights from the UX domain explored user feedback on how VQ and FoV influenced their depth perception experience.

- *Objective Data:* ADED results showed a marginally better depth estimation under WideFoV conditions compared to NarrowFoV, although the differences were not statistically significant. The consistent trends across conditions suggest that while FoV influences spatial awareness, the effect size remains small. Similarly, variations in VQ had minimal impact on ADED, indicating that users were able to perform depth estimation tasks effectively across different video quality levels.
- *Quantitative Insights:* Participants rated the impact of FoV on depth perception higher than VQ, with WideFoV enabling more consistent and confident depth judgments. The results also showed variability in users' confidence ratings under NarrowFoV and lower VQ conditions. This indicates that reduced visual context (NarrowFoV) or poor clarity (low VQ) affected users' confidence in depth estimation [83, 84], even when their objective performance (ADED) remained relatively stable.
- *Qualitative Feedback:* Participants frequently reported that WideFoV enhanced their spatial awareness and depth perception, as it aligned more closely with natural human vision. However, some noted distortions, such as the "fish-eye effect" under WideFoV, could complicate depth estimation. Conversely, NarrowFoV limited visible context, making it harder for users to judge distances accurately, particularly under high-latency conditions. Regarding VQ, participants generally found the provided video quality sufficient for depth estimation but noted that higher VQ improved their confidence and reduced visual strain.

In summary, while the objective and quantitative results indicate minimal differences in performance across VQ and FoV conditions, the qualitative insights emphasise the importance of user confidence and perceived usability. WideFoV emerges as a preferred configuration for supporting depth perception, but care should be taken to minimise distortions. Improvements in video quality, though not critical for objective performance, enhance user confidence and overall experience.

RQ3: Influence of VQ, FoV, and latency on UX and QoE

The impact of VQ, FoV, and latency on the General user experience and Quality of Experience in the teleoperation highlights critical factors that shaped user perceptions and task performance. These insights are synthesised below:

Users' Comfort and Interaction Experience: The quantitative results demonstrate that configurations with low video quality, high latency, and narrow field of view significantly reduced users' comfort and interaction experience ratings. Notably, T18 (*LoQual_HiLat_NarrowFoV*) consistently received the lowest scores, with post-hoc comparisons confirming its significant deviation from high-quality configurations like T1 (*HiQual_LoLat_WideFoV*).

The moderate effect size ($\eta_g^2 = 0.082$) suggests that these visual degradations notably impacted user comfort. This indicates that ensuring high-quality video and minimising latency is vital for fostering a positive interaction experience in teleoperation tasks.

Simulator Sickness: While simulator sickness symptoms were generally mild, a few participants experienced visual strain. This finding highlights the importance of addressing oculomotor strain in teleoperation design by improving video clarity, refresh rates, and latency to reduce visual fatigue. By maintaining tolerable levels of discomfort, designers can ensure the system remains usable across extended sessions.

Qualitative feedback on system usability and improvements: Participants emphasised the need to reduce latency, improve video quality, and provide wider or additional camera views to enhance usability. Latency was identified as a key factor disrupting smooth control and causing cognitive strain, particularly in high-latency conditions. Suggestions for integrating AR/VR technologies, realism-enhancing features (e.g., steering wheel-like controls), and intuitive visual aids (e.g., virtual guidelines) highlight the potential for innovative solutions to bridge the gap between teleoperation and real-world experiences. These enhancements could improve users' confidence, engagement, and overall satisfaction with the system [40].

Hedonic Aspects of UX: Hedonic aspects showed that users enjoyed the novelty and interactivity of the system, with many describing the experience as fun and engaging.

However, there were also challenges in adapting to latency, depth perception issues, and technical limitations, which showed areas for improvement. Despite the challenges, participants demonstrated adaptability. This balance between enjoyment and challenge underscores the importance of designing systems that are both intuitive and capable of fostering skill development [58].

RQ4: Enhancing the understanding of user interaction in teleoperation through a mixed-methodology approach?

Integrating mixed-method approaches in this study provided a comprehensive perspective on user interaction within the mimicked teleoperation system. This approach includes measurable outcomes and the subjective nuances of QUX. These insights are described below:

Holistic Understanding Through Mixed Data: Including objective performance metrics, such as ADED and truck deviation, provided precise and measurable insights into task performance. These metrics demonstrated clear interactions between visual configurations and latency, highlighting how latency and FoV combined to influence performance. However, these findings alone could not capture the experiential and perceptual challenges users faced. Quantitative QoE measures, such as system responsiveness and comfort ratings, complemented the objective data by revealing users' perceptions of system performance under different configurations, including the significant impact of latency and Narrow FoV on perceived responsiveness and comfort. Finally, qualitative feedback provided the narrative context, elucidating how users experienced cognitive strain, adapted to latency, or relied on multisensory feedback to mitigate challenges. This triangulation of data sources enriched the findings by offering depth and breadth [60].

Contextualising User Perception and Performance: While objective metrics showed minimal differences in certain conditions, such as ADED across VQ levels, qualitative feedback revealed nuanced user experiences, including reduced confidence under low VQ or Narrow FoV conditions. This alignment between performance metrics and user perceptions underscores the importance of mixed methodologies in highlighting how users interact with the system beyond measurable outcomes. For example, the misattribution of latency to the game controller (despite technical measurements indicating visual feedback delays) demonstrates how qualitative insights can contextualise quantitative findings, enabling researchers to identify gaps between user perception and technical realities.

Implications for teleoperation Design and Research: The mixed-method approach provided a comprehensive evaluation of user interaction and informed actionable design improvements. Qualitative insights, such as preferences for wider FoV, higher video quality, and latency reduction, were supported by quantitative findings showing significant differences in comfort and interaction experience under these configurations. Moreover, participants' feedback on integrating AR/VR technologies and multisensory feedback highlights avenues for improving the intuitiveness and usability of teleoperation systems.

By addressing both the technical and perceptual dimensions of user interaction, mixed methodologies guided us towards a more user-centred design process. The mixed-method approach proved to deepen the understanding of user interaction in teleoperation. It provided a robust framework to holistically examine performance, perception, and experience, enabling nuanced insights that guide both practical improvements and future research directions.

Limitations of the study

Reflection on Gender Representation in the Study Sample: This study included 65% male participants, reflecting a gender imbalance that may mirror trends in industrial teleoperation but could also be attributed to recruitment limitations. Such a sample limits the generalisability of findings, especially regarding how diverse user groups might perceive or interact with remote systems. We also acknowledge potential WEIRD (Western, Educated, Industrialised, Rich, and Democratic) biases in the sample, which may further restrict cross-cultural applicability. Future research should adopt more deliberate sampling strategies to better represent user diversity and support the design of inclusive, adaptable remote operation systems.

Reflections on Research Positionality: This study adopts a user-centred, mixed-methods approach informed by both HCI and Telecommunications traditions. As discussed earlier in Sections "Integration of the QoE and UX in teleoperation" and "Knowledge gap", UX and QoE are treated as complementary perspectives on user interaction. Here, we reiterate that the distinction applied served primarily to structure data analysis rather than to assert a theoretical divide. QoE was operationalised through ratings and performance metrics, while UX was explored via qualitative insights. This reflects recent work acknowledging the convergence of these constructs [23, 24, 42, 46, 48, 49].

Reflections on Ecological Validity: We recognise that some elements of the setup–such as visual floor markings and audible engine sound–may have introduced artificial cues not typical of all remote operation scenarios, potentially affecting depth perception and control. However, these cues reflect features common in real-world contexts (e.g., construction sites) and unintentionally offer insights into the role of incidental sensory information. Additionally, as

noted in Sec. 3.2.1, lighting was calibrated to approximate daylight, but the lab could not replicate outdoor luminance, which may have subtly influenced visual perception; future studies should explore more realistic conditions to enhance ecological validity.

Methodological Reflections: This study used open-ended questionnaires to capture diverse user experiences, though the absence of interviews limited the depth of qualitative insights. Thematic analysis followed a two-phase coding process, involving grounded coding and thematic structuring, as outlined in Braun and Clarke's framework [78]. This process was conducted by a single coder with support from Dovetail software. Future work should involve multiple coders to enhance analytical rigour. Regarding sample size, simulator sickness, and session duration, the study followed ITU-R BT.500 guidelines [85] with 20 participants and used SSQ [66], which was appropriate for the setup; however, future studies should plan a priori power analysis (conducting power analysis based on the observed variance in the current study, during the planning phase for the selection of target sample size), consider VR-specific alternatives like VRSQ or ARSQ, and include formal fatigue tracking. Despite the 75-minute average session duration, no fatigue-related complaints were reported, likely due to implemented mitigation strategies; nonetheless, future studies should include formal fatigue tracking.

Conclusion

This study provides critical insights into the interplay of video quality, field of view, and latency in shaping performance, depth perception, and user experience in remote-control systems designed for industrial applications. This research utilised a user-centred, mixed-method approach, considering Quality of Experience and User Experience insights through a controlled experimental laboratory investigation. It examined users' performance, depth perception, and overall user experience under constrained visual conditions, addressing challenges commonly encountered in real-world teleoperation tasks. The findings show that latency has the most significant impact on user performance, followed by field of view. In particular, the narrow field of view is sensitive to high latency, possibly due to its limited spatial context, which makes it harder for users to compensate for delayed system responses. The field of view was more critical in perceiving depth than video quality in this experimental setup. Users perceive wide field of view as beneficial for depth perception, but field-of-view-related distortions need to be considered. While video quality had a limited impact on depth estimation accuracy (as shown in quantitative data), it enhances user confidence and reduces visual strain (as indicated by qualitative feedback). These findings highlight the importance of considering user experience in teleoperation design, reinforcing that an optimised field of view and higher video quality contribute to better user comfort and confidence, even when objective performance remains stable. Multi-sensory feedback, such as audio cues, emerged as a valuable enhancement to user interaction, helping to bridge gaps caused by latency and visual limitations. These results underscore the need for future designs to prioritise adaptive visual configurations, latency optimisation, multi-sensory integration, and the incorporation of technologies such as augmented reality and haptic feedback to support intuitive user interaction.

Acknowledgements This research has been funded by the Swedish Foundation for Strategic Research (SSF dnr. FID18-0030) and also by Sweden´s Innovation Agency (VINNOVA, dnr. 2021-02107) through the Celtic-Next project IMMINENCE (C2020/2-2, dnr. 2021-02107), SCREENS II (VINNOVA, dnr. 2023-00755), and Control (VINNOVA, dnr. 2022-02670). The authors also acknowledge the reviewers and their comments, which have greatly enabled us to improve the manuscript.

Funding Open access funding provided by Mid Sweden University.

Declarations

Conflict of interest On behalf of all authors, the corresponding author states that there is no Conflict of interest.

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